Search for axionlike particle as dark matter candidate with nuclear magnetic resonance

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Declaration of Authorship

I hereby declare that this thesis titled, *Search for axionlike particle as dark matter candidate with nuclear magnetic resonance* and the work presented in it are my own. I confirm that:

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- Where I have consulted the published work of others, this is always clearly attributed.
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Notes

To avoid confusions, constants such as light speed and Planck constant are explicitly written in this thesis. Transformation to different unit systems can be made by choosing the values of the constants.

SI unit system is adopted to be compatible with experimental results.

The source of this dissertation, including LaTeX files, program scripts, and figures, is available in the GitHub repository of the CASPEr Collaboration:

https://github.com/CASPEr-Collaboration/ZhangYuzhe-Dissertation .

Yuzhe: Add data availability statement: specify DOI or repository where the experimental data are deposited.

Abstract

Axions and axionlike particles (ALPs) are theoretically well-motivated candidates for the dark matter that constitutes approximately 27 % of the total energy content of the universe. Through the nuclear gradient coupling, an oscillating ALP dark-matter field acts as an effective pseudomagnetic field on nuclear spins, driving resonant spin precession at the ALP Compton frequency. Nuclear magnetic resonance (NMR) is among the most sensitive techniques for detecting weak oscillating fields, making it a natural tool for direct ALP dark-matter searches.

This thesis presents a theoretical and experimental investigation of ALP dark-matter searches using precision NMR in the context of the Cosmic Axion Spin Precession Experiment (CASPEr). On the theoretical side, the signal amplitude and optimal measurement strategy for a frequency-scanning search are derived analytically. Depending on the relative magnitudes of the NMR coherence time T_2^*, the intrinsic transverse relaxation time T_2, and the ALP coherence time τ_a, four distinct signal regimes are identified, each with a characteristic sensitivity and optimal dwell-time allocation. In all regimes, the sensitivity to the ALP-nucleon coupling strength scales as $T_{\mathrm{tot}}^{-1/4}$ with total measurement time, motivating improvements in spin polarization and relaxation times over simply accumulating more data. The optimal frequency step is equal to the sensitive bandwidth of a single measurement, and the dwell time should be distributed uniformly across frequency steps in regimes (1)–(3), or proportional to ν^{-1} in regime (4).

On the experimental side, the scanning framework is applied in a proof-of-concept search for the ALP-proton gradient coupling using the CASPEr-Gradient-Low-Field apparatus at the Johannes Gutenberg University Mainz. A thermally polarized liquid methanol sample is used to scan a 240 Hz band centered at 1.348570 MHz (proton Larmor frequency), corresponding to ALP masses in the range 5.576741–5.577733 neV/c^2. Three sequential scans were performed; no ALP signal candidates survived a cross-scan consistency check. A 95 % confidence-level upper limit of $|g_{\mathrm{ap}}| \lesssim 3 \times 10^{-2}\,\mathrm{GeV}^{-1}$ is set on the ALP-proton gradient coupling in this mass range, constituting the first dedicated laboratory search therein.

The sensitivity of the current measurement is dominated by the small thermal polarization of the methanol sample ($\approx 1.8 \times 10^{-7}$). Projected sensitivity estimates demonstrate that hyperpolarized samples—parahydrogen-induced polarization, dynamic nuclear polarization, or spin-exchange optical pumping of liquid

^{129}Xe—can improve the coupling sensitivity by several orders of magnitude, opening a path toward new physics beyond current bounds.

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1 Introduction

The central observation motivating this thesis is that axionlike particles (ALPs) interact with nuclear spins as an effective oscillating pseudomagnetic field. Nuclear magnetic resonance (NMR) is among the most sensitive techniques for detecting weak oscillating fields, making it a natural tool for direct ALP dark-matter searches.

Astrophysical evidence for dark matter comes from galaxy rotation curves, gravitational lensing, large-scale structure, and the cosmic microwave background: visible matter alone cannot account for the observed dynamics. Axions and axionlike particles are theoretically attractive candidates because they are weakly coupled, naturally light, and can generate coherent spin-precession signals accessible to precision NMR experiments.

This chapter introduces the key ingredients of the search. Section 1.1 reviews the NMR concepts most relevant to the experiment. Section 1.2 summarizes the dark-matter evidence. Section 1.3 describes the axion and ALP framework, including the pseudomagnetic-field analogy that connects ALP physics to NMR. Section 1.4 outlines the CASPEr experimental program. The theory of signal estimation and scanning optimization is developed in Chapter 2; the apparatus is described in Chapter 3; measurements and results appear in Chapter 4.

1.1 Nuclear magnetic resonance

Nuclear magnetic resonance is a technique for studying nuclear spins in magnetic fields. In the presence of a leading field, spin states split by the Zeeman effect and a polarized sample acquires a net magnetization that precesses at the Larmor frequency. Because the resonant frequency is directly proportional to the magnetic field, NMR is a precision tool for field measurement.

The properties most relevant to this thesis are sensitivity to weak fields, applicability from zero to very high field, and a rich time-domain response. These allow axionlike spin precession to be treated as a small perturbation on a standard NMR experiment.

1.1.1 Zeeman splitting and polarization

In a magnetic field B_0, a spin-1/2 nucleus splits into two energy levels separated by

$$\Delta E = \hbar\gamma B_0\,, \tag{1.1}$$

where γ is the nuclear gyromagnetic ratio. This energy difference sets the Larmor (resonance) frequency

$$\nu_L = \frac{\gamma B_0}{2\pi}\,. \tag{1.2}$$

For protons, $\gamma/(2\pi) = 42.577\,\mathrm{MHz/T}$, so a bias field of 31.67 mT (as used in this work) gives $\nu_L \approx 1.348\,\mathrm{MHz}$.

At thermal equilibrium, the spin populations obey the Boltzmann distribution. In the high-temperature limit $\Delta E \ll k_B T$ (satisfied for all conditions in this thesis), the fractional spin polarization is

$$p \approx \frac{\hbar\gamma B_0}{2k_B T}\,. \tag{1.3}$$

At $B_0 = 31.67\,\mathrm{mT}$ and $T = 180\,\mathrm{K}$, the proton thermal polarization is only $p \approx 1.8 \times 10^{-7}$. Hyperpolarization techniques—such as spin-exchange optical pumping or parahydrogen-induced polarization—can boost p by several orders of magnitude, which is the main route to improved sensitivity identified in this thesis.

1.1.2 Bloch equations

The evolution of the nuclear spin magnetization $\mathbf{M}$ in the combined bias and pseudomagnetic fields is governed by the Bloch equations [2]:

$$\frac{dM_x}{dt} = \gamma(\mathbf{M}\times\mathbf{B}_{\text{eff}})_x - \frac{M_x}{T_2}\,, \tag{1.4}$$

$$\frac{dM_y}{dt} = \gamma(\mathbf{M}\times\mathbf{B}_{\text{eff}})_y - \frac{M_y}{T_2}\,, \tag{1.5}$$

$$\frac{dM_z}{dt} = \gamma(\mathbf{M}\times\mathbf{B}_{\text{eff}})_z - \frac{M_z - M_0}{T_1}\,, \tag{1.6}$$

where $\mathbf{B}_{\text{eff}} = \mathbf{B}_0 + \mathbf{B}_a(t)$ is the total effective field, M_0 is the equilibrium magnetization, and T_1, T_2 are the longitudinal and transverse relaxation times. Because $|\mathbf{B}_0| \gg |\mathbf{B}_a|$ by many orders of magnitude, the equations are most efficiently solved in the rotating coordinate frame (RCF), which removes the fast Larmor precession at ν_L and exposes the much slower ALP-induced tipping dynamics.

In the RCF, when the Larmor frequency is tuned to the ALP Compton frequency ($\nu_L = \nu_a$), the transverse component of $\mathbf{B}_a$ acts as a near-static drive. In the weak-drive limit ($\Omega_a T \ll 1$), the magnetization accumulates a small tipping angle ξ and the transverse magnetization is $M_1 \approx M_0 \xi$. The ALP field is stochastic (Section 2.1.3), so ξ and M_1 must be treated statistically; their r.m.s. values are derived in Section 2.1.4.

1.1.3 Spin-projection noise

Even in the absence of a leading field, an ensemble of N spins does not cancel perfectly. Statistical fluctuations leave a residual transverse moment of order $\sqrt{N}\,\mu$ (where $\mu = \hbar\gamma/2$ is the nuclear magneton), giving a magnetization noise floor

$$M_{\text{SPN}} \sim \frac{\sqrt{N}\,\mu}{V}\,. \tag{1.7}$$

This spin-projection noise (SPN) is the fundamental quantum limit for spin measurements [21]. In the axion-search context it sets the minimum detectable transverse magnetization and is listed in Table 3.2 for the experimental parameters of this work.

1.1.4 Relaxations

Relaxation describes the return of the spin ensemble toward equilibrium after it has been perturbed. Longitudinal relaxation T_1 controls how quickly the magnetization along the leading field rebuilds, while transverse relaxation determines how quickly coherent precession dephases. Two transverse relaxation times must be distinguished:

$$\frac{1}{T_2^*} = \frac{1}{T_2} + \frac{1}{T_\delta}\,, \tag{1.8}$$

where T_2 characterizes irreversible spin-spin interactions and T_δ quantifies the additional dephasing caused by magnetic-field inhomogeneity across the sample volume. Because field inhomogeneity can be reduced by shimming (Chapter 3), T_2^* is experimentally controllable, but T_2 sets an intrinsic upper bound.

These time scales are central to the thesis because they set the window over which the ALP-induced signal can be observed coherently. Longer T_2^* improves sensitivity in all four signal regimes derived in Chapter 2.

T_1 relaxation

Longitudinal relaxation governs how the population difference between spin states rebuilds after perturbation. In practice, a short T_1 is helpful for rapid repetition, while a long T_1 can preserve polarization for longer measurements.

T_2^* relaxation

Transverse relaxation combines microscopic spin-spin interactions (T_2) and macroscopic field inhomogeneity (T_δ) as given in Equation (1.8). It determines the free-decay[1] signal and directly controls the linewidth in the frequency domain. For field searches, a narrower linewidth can improve sensitivity, but only if the coherence time remains long enough to integrate signal power efficiently.

Cross relaxation

Cross relaxation transfers energy between coupled spin species or spin reservoirs. In samples containing more than one spin system, it modifies the effective relaxation rates of each species.

1.1.5 Chemical shift

Chemical shielding modifies the local field at a nucleus and shifts its resonance frequency. The shift is small in absolute terms but distinguishes nuclei in different chemical environments.

In this thesis, chemical shift contributes to the spectral structure of the sample and offsets the resonance from the bare Larmor value.

1.1.6 Spin-spin coupling

Coupling between nuclear spins splits resonance lines into multiplets. The resulting J-coupling pattern provides a spectral fingerprint for molecular identification, which helps distinguish genuine NMR signals from noise.

For the calibration sample, J-coupling produces a resolved multiplet that serves as a reliable spectral reference.

1 Yuzhe: remove induction and mention there might not be any induction

1.1.7 Experimental schemes

Two pulse-based schemes are especially useful. Free induction decay measures the coherent response after a tipping pulse, while spin echo refocuses dephasing and gives access to the intrinsic transverse relaxation time. Both are diagnostic tools that help characterize the apparatus before an axion measurement.

Free decay

After a $\pi/2$ pulse, the magnetization is tipped into the transverse plane and then decays with the dephasing time T_2^*. The resulting free-induction signal is the simplest way to measure resonance frequency and linewidth.

Spin echo

Spin echo uses a π pulse to reverse dephasing from static field inhomogeneity. By recovering the signal at the echo time, it separates true spin-spin relaxation from broadening caused by imperfect field homogeneity, yielding a direct measurement of T_2.

CPMG sequence

The Carr–Purcell–Meiboom–Gill (CPMG) sequence extends the spin-echo scheme by applying a $\pi/2$ pulse followed by a train of π refocusing pulses. Each refocusing pulse produces an echo, and the decay of the echo amplitudes gives a robust measurement of T_2. In this thesis, CPMG is used as a pulsed-NMR diagnostic to characterize the sample relaxation before and after long ALP-search measurements.

1.2 Dark matter hypothesis

The concept of dark matter arose from astronomical observations that could not be explained by visible matter and standard gravity alone. Galaxy rotation curves are the classic example: the outer parts of galaxies rotate too quickly for the observed baryonic mass to provide the required gravitational binding [25, 26]. The Bullet Cluster provides direct evidence through the spatial separation between the gravitational mass (traced by lensing) and the baryonic mass (X-ray gas) after two galaxy clusters collide [12]. The cosmic microwave background power spectrum and big-bang nucleosynthesis abundances independently constrain baryonic matter to roughly 5 % of the total energy budget, with dark matter contributing about 27 % [3].

1.2.1 History

Evidence for dark matter accumulated from galaxy-dynamics studies, gravitational lensing, and large-scale structure surveys. Each line of evidence independently supports a non-baryonic matter component that dominates the mass budget of the universe.

1.2.2 Recent developments

Modern dark-matter searches span weakly interacting massive particles (WIMPs), ultralight bosons, and direct-detection strategies based on precision measurement [19]. WIMPs have been strongly constrained by direct-detection experiments; axions and axionlike particles have attracted increasing attention because they connect cosmology with laboratory-scale experiments through tiny but coherent effects.

1.3 Axions and axionlike particles

The QCD axion was originally proposed in 1977 by Peccei and Quinn as a solution to the strong *CP* problem [22, 23]. A global symmetry breaking at the scale f_a generates a pseudo-Goldstone boson whose potential is minimized at the *CP*-conserving point, naturally explaining why the strong interaction conserves *CP*. The axion acquires a mass $m_a \sim \Lambda_{\mathrm{QCD}}^2/f_a$ from QCD dynamics.

1.3.1 QCD axion

The QCD axion is a hypothetical pseudoscalar particle that appears when a global symmetry is introduced to explain why strong interactions conserve *CP* so well. Its properties are tightly linked to the QCD scale, which makes it a special case rather than the most general one. Cosmological production via the misalignment mechanism constrains $f_a \lesssim 10^{12}$ GeV (axion mass $\gtrsim 6\,\mu\mathrm{eV}$) if the axion is to constitute all dark matter [1, 13, 24].

1.3.2 Axionlike particles

Axionlike particles share the oscillatory and weakly coupled character of the QCD axion, but their mass range can be much wider. This flexibility makes them attractive targets for laboratory searches.

On time scales shorter than the ALP coherence time, the ALP field can be treated as a classical oscillation:

$$a(\mathbf{r}, t) = a_0 \cos(\nu_a t - \mathbf{k}\cdot\mathbf{r} + \phi)\,, \tag{1.9}$$

where $\nu_a = m_a c^2/\hbar$ is the ALP Compton frequency, $\mathbf{k} = m_a \mathbf{v}_a/\hbar$ is the wave vector, a_0 is the amplitude, and ϕ is a random phase. The amplitude a_0 is set by the local dark-matter density through $\cancel{\rho_{DM} \approx c\, m_a^2 a_0^2/(2\hbar^3)}$Yuzhe: comment [11]. Because the ALP number density is so high for ultralight masses, this classical-field description is appropriate.

Quantum nature

1.3.3 Non-gravitational interactions

The most relevant non-gravitational couplings for this thesis are the gradient coupling to nucleons. The nucleon Hamiltonian in the presence of an ALP field background can be written as [17]

$$H_{\mathrm{aNN}} = c g_{\mathrm{aNN}}\, \boldsymbol{\nabla} a \cdot \mathbf{I}\,, \tag{1.10}$$

where g_{aNN} is the coupling strength in GeV^{-1} and $\mathbf{I}$ is the nuclear spin operator. In this form of Hamiltonian, $g_{\mathrm{aNN}} a_0$ becomes dimensionless. The r.m.s. value of ALP field amplitude is set by the local dark-matter energy density [8, 20, 26]:

$$a_0 = \sqrt{\frac{2\rho_{DM}\hbar^3}{m_a^2 c}}\,. \tag{1.11}$$

The amplitude a_0 here has dimension of energy.

In some papers, nuclear spin operator $\mathbf{I}$ is substituted with $\hbar\boldsymbol{\sigma}_n$, where $\boldsymbol{\sigma}_n$ is dimensionless nuclear spin operator. Expanded:

$$H_{\mathrm{aNN}} = c g_{\mathrm{aNN}} a_0 \sin(\nu_a t - \mathbf{k}\cdot\mathbf{r} + \phi) m_a\, \mathbf{v}_a \cdot \mathbf{I}/\hbar\,, \tag{1.12}$$

Comparing this with the NMR Hamiltonian $H_{\mathrm{NMR}} = \gamma \mathbf{B}\cdot\mathbf{I}$, the ALP gradient acts as an effective oscillating pseudomagnetic field

$$\mathbf{B}_a(t) = \frac{c g_{\mathrm{aNN}}\, \boldsymbol{\nabla} a}{\gamma}\,. \tag{1.13}$$

This gradient coupling oscillates at the ALP Compton frequency and can drive resonant spin precession when the Larmor frequency matches ν_a. The search for this coupling is the subject of Chapter 4. Yuzhe: add some examples to show equivalent Ba field magnitude?

The r.m.s. Rabi frequency of the pseudomagnetic field can be found by the Hamiltonian in Equation (1.12) [11, 32]

$$\Omega_a = \frac{1}{2\hbar} c g_{\text{aNN}} a_0 m_a v_\perp = \frac{1}{2} g_{\text{aNN}} \sqrt{2\hbar c \rho_{DM}} v_\perp , \tag{1.14}$$

where $v_\perp$ is the component of the ALP-detector relative velocity perpendicular to $\mathbf{B}_0$. Taking $v_\perp = 220\,\text{km}\,\text{s}^{-1}$, we have $\Omega_a/2\pi = g_{\text{aNN}} \times 0.2\,\text{GeV} \cdot \text{Hz}$. For $g_{\text{aNN}} \lesssim 10^{-10}\,\text{GeV}^{-1}$, the Rabi period $2\pi/\Omega_a \gtrsim 1000\,\text{yr}$, confirming that the ALP field is a weak drive on the spins. Yuzhe: about that one half? btw check the values! Mention weak-drive limit

1.4 CASPEr project

Yuzhe: first mention a lot of magnetic resonance searches (haloscope). [19]

CASPEr (Cosmic Axion Spin Precession Experiment) is a family of experiments proposed in 2014 [9] that use nuclear spins to search for axion-induced effects. The idea is to translate the tiny axion coupling in Equation (1.13) into a measurable spin precession in a carefully controlled NMR setup.

The project has several branches, including electric and gradient-based searches, and both low-field and high-field implementations. The low-field regime is especially relevant when the goal is to scan a wide frequency band with high sensitivity to weak couplings.

1.4.1 Idea

The central idea is to tune the NMR resonance to a frequency that matches the axion Compton frequency. If the two frequencies align, the axion field acts as the pseudomagnetic field $\mathbf{B}_a(t)$ in Equation (1.13) and induces a coherent transverse magnetization in the sample.

In practice, the method reduces to a frequency scan: either the signal appears at a particular resonance or the absence of a signal constrains the coupling strength over the scanned band.

1.4.2 CASPEr-Electric

Yuzhe: shoten these subsections CASPEr-Electric (CASPEr-E) searches for an oscillating nuclear electric dipole moment induced by the ALP-gluon coupling. This moment generates an effective oscillating field along the nuclear spin axis in the presence of an electric field, enabling detection through NMR at the corresponding Compton frequency.

1.4.3 CASPEr comagnetometers

Comagnetometer configurations use two nuclear spin species in the same volume to suppress common-mode noise from magnetic-field fluctuations. The differential signal between the two species enhances sensitivity to a pseudomagnetic ALP field while rejecting electromagnetic backgrounds. Early results were obtained with a liquid-state comagnetometer [29] and an ultralow-field configuration [16].

1.4.4 CASPEr-Gradient

CASPEr-Gradient searches for the ALP-fermion gradient coupling described by Equation (1.10). Several setups at Mainz cover different ALP-mass ranges. The CASPEr-Gradient-Low-Field experiment uses bias fields up to about 0.1 T, corresponding to proton Larmor frequencies up to approximately 4.3 MHz. The CASPEr-Gradient-High-Field experiment will use a tunable 14.1 T magnet to reach frequencies up to 600 MHz. A conceptual diagram of the low-field apparatus is shown in Figure 1.1. The cryostat contains a liquid helium reservoir, excitation and pickup coils, and a SQUID magnetometer. A superconducting magnet produces the bias field $\mathbf{B}_0$, which sets the Larmor frequency and can be varied to scan ALP masses.

The resonance scheme is the main search strategy for CASPEr-Gradient-Low-Field. The bias field is stepped through a range of values, and the NMR response is recorded at each step. A resonant signal appears as a peak in the transverse magnetization when ν_L matches ν_a. The broadband scheme, which is more suitable for higher frequencies, continuously monitors the NMR response without stepping the bias field, looking for transient signals that could indicate an ALP interaction.

1.4.5 Scientific targets beyond gradient coupling

Yuzhe: mention what are going to be included? in the coming chapters?

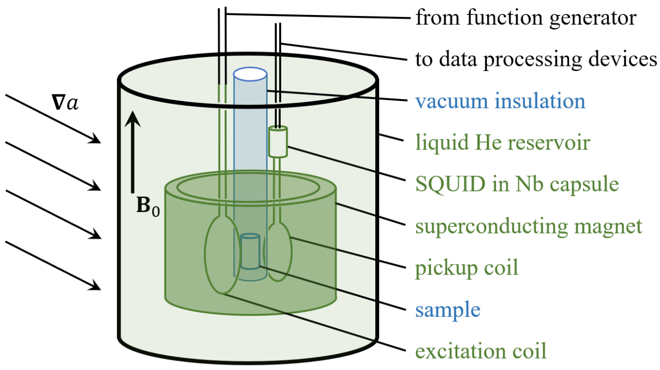

Figure 1.1: Conceptual diagram of the CASPEr-Gradient-Low-Field setup. Superconducting devices (dark green) are submerged in a liquid helium bath (light green). The excitation and pickup coils characterize the sample and detect the spin-precession signal. The sample temperature can be controlled independently of the helium bath, enabling measurements with both thermal and hyperpolarized samples. Reproduced from Ref. [32].

2 Axionlike-particle-induced spin precession

Part of the content in this chapter related to frequency-scanning considerations is adapted from Ref. [32]. Yuzhe: modify?

The ALP field can be detected through its gradient coupling to nuclear spins, which acts as an effective oscillating pseudomagnetic field. Understanding the properties of this coupling—its frequency, amplitude, and coherence—is essential for designing a sensitive experiment and choosing an optimal measurement strategy. This chapter builds the signal model from first principles: starting from the quantum-mechanical description of a single ALP and showing how the superposition of many such quanta gives rise to a classical stochastic oscillating field, which then drives resonant spin precession in the NMR sample.

2.1 The ALP dark-matter field

2.1.1 Single particle

Non-relativistic ALP with velocity $\mathbf{v}_a \ll c$ can be described by a plane-wave state

$$a(\mathbf{r}, t) = a_0\, e^{\,i(\mathbf{k}\cdot\mathbf{r} - 2\pi\nu_a t)}\,, \tag{2.1}$$

where the Compton frequency $\nu_a = m_a c^2/h$ is set by the rest mass alone, and the wavevector $\mathbf{k} = m_a \mathbf{v}_a/\hbar$ carries the kinetic information. The de Broglie wavelength of such a particle is

$$\lambda_{dB} = \frac{h}{m_a v_a}\,. \tag{2.2}$$

For $m_a < 2.5\,\mu\mathrm{eV}$Yuzhe: mention CASPEr scan range and $v_a < 550\,\mathrm{km\,s^{-1}}$[15], this gives $\lambda_{dB} > 1.7\,\mathrm{km}$Yuzhe: check the numbers here. add a reference to calculation script. The wavelength grows as m_a^{-1}, so at the neV masses probed in this thesis it reaches hundreds of kilometres—far larger than any laboratory apparatus. This means the ALP field is spatially coherent across the entire experiment, and the spatial gradient $\boldsymbol{\nabla} a$ that couples to nuclear spins is well approximated as uniform over the sample volume.

The enormous de Broglie wavelength also implies an exceptionally high phase-space occupation number. The number density of ALPs in the local dark-matter halo is

$$n_a = \frac{\rho_{DM}}{m_a c^2} \approx \frac{0.4\,\mathrm{GeV/cm^3}}{m_a c^2}\,, \tag{2.3}$$

so the mean occupation number per de Broglie mode is

$$\mathcal{N} \approx n_a\,\lambda_{dB}^3 = \frac{\rho_{DM}}{m_a c^2}\left(\frac{hc}{m_a c^2}\right)^3\left(\frac{c}{v_a}\right)^3. \tag{2.4}$$

For $m_a = 1\,\mu\mathrm{eV}$ and $\rho_{DM} = 0.4\,\mathrm{GeV/cm^3}$, one obtains $\mathcal{N} \approx 10^{30}$. Even for the lightest masses still in the particle regime ($m_a \sim 10^{-22}$ eV, "fuzzy" dark matter), $\mathcal{N}$ is astronomically large. This is a direct consequence of the tiny ALP mass: for a fixed energy density, a lighter boson must be exponentially more abundant.

2.1.2 Ensemble

Because ALPs are bosons and their occupation numbers are so large, quantum statistics strongly suppress number fluctuations relative to the mean: $\delta\mathcal{N}/\mathcal{N} \sim \mathcal{N}^{-1/2} \to 0$. In this regime, the quantum field operator $\hat{a}$ can be replaced by a classical field. Each ALP contributes a plane wave with amplitude $a_i = a_0/\sqrt{N_a}$, wavevector $\mathbf{k}_i = m_a \mathbf{v}_i/\hbar$, and a random phase ϕ_i:

$$a(\mathbf{r}, t) = \frac{a_0}{\sqrt{N_a}} \sum_i \cos(\nu_i t - \mathbf{k}_i \cdot \mathbf{r} + \phi_i)\,, \tag{2.5}$$

where $\nu_i = \nu_a + \frac{1}{2} m_a v_i^2/h$ is the total energy (including kinetic energy). Note that the ALP amplitude here is normalized by $\sum_i a_i^2 = a_0^2$.

2.1.3 Stochasticity

The modes in Equation (2.5) have slightly different frequencies due to the ALP ensemble velocity spread. Their phases drift apart over time, hence decoherence happens. If the velocity spread, characterized by FWHM of the velocity distribution, is Δv_a, the corresponding kinetic-energy spread would be $\Delta E_k = \frac{1}{2} m_a (\Delta v_a)^2$. This results in a relative spread of ALP Compton frequencies of

$$\Delta\nu_a/\nu_a = \frac{\Delta E_k}{h\nu_a} = \left(\frac{\Delta v_a}{c}\right)^2. \tag{2.6}$$

The ALP quality factor Q_a is defined as the inverse of this relative spread as $Q_a = \nu_a/\Delta\nu_a$. The quality factor here estimates the number of oscillations after which the phases of ALPs diverge. A rough estimation can be made considering two ALPs different in frequency by a factor of $\Delta\nu_a$. After Q_a oscillations (so-called coherence time $\tau_a \approx Q_a/\nu_a = 1/\Delta\nu_a$), the phase difference accumulated is approximately $2\pi\Delta\nu_a\tau_a = 2\pi$, which means these two ALPs are no longer coherent.

On longer time scales the random phases between modes cause the amplitude and phase of the effective oscillation to drift, so the field must be treated as a stochastic process [11, 18]. For Compton frequencies above 1 kHz, $\tau_a \lesssim 1000$ s; the interplay between τ_a and the NMR relaxation times determines the optimal measurement strategy in Chapter 4. Yuzhe: here needs more input

2.1.4 Spin-precession signal

The ALP-induced signal is determined by the ALP gradient field amplitude (or effectively, the Rabi frequency of the pseudomagnetic field Ω_a), the ALP field coherence and relaxations of the nuclear spin sample. An estimation of the effects from these factors is made in the following.

Relaxations and coherence

Yuzhe: include alp COHERENCE AS well The transverse spin relaxation in the sample arises from homogeneous mechanisms (e.g., spin-spin interactions), characterized by the time T_2, and from inhomogeneous mechanisms (e.g., magnetic-field gradients across the sample), which cause additional dephasing. The total effective transverse relaxation time $T_2^* \le T_2$ incorporates both contributions.

The NMR linewidth (FWHM, assuming a Lorentzian lineshape) of the free-induction-decay spectrum is

$$\Gamma_n = \frac{2}{T_2^*}, \tag{2.7}$$

and the normalized linewidth is

$$\delta_\nu \equiv \frac{\Gamma_n}{\nu} = \frac{2}{\nu T_2^*}, \tag{2.8}$$

often expressed in parts per million (ppm). The inhomogeneous contribution to $1/T_2^*$ can be reduced by magnetic shimming, giving control over T_2^* independently of the intrinsic T_2.

Estimation

For a sample with longitudinal magnetization M_0 prepared along the bias field $\mathbf{B}_0$, the ALP-field drive tilts the magnetization by a small angle $\xi \ll 1$ (weak-drive regime, $\Omega_a T \ll 1$), generating a transverse magnetization $M_1 \approx M_0\xi$. The r.m.s. transverse magnetization is [32]

$$\sqrt{\langle M_1^2 \rangle} \approx \frac{1}{\sqrt{2}} u_n M_0 \sqrt{\langle \xi^2 \rangle}\,, \tag{2.9}$$

where u_n is the fraction of nuclear spins on resonance with the ALP field. Approximating both the NMR and ALP lineshapes as rectangles,

$$u_n \approx \begin{cases} 1\,, & \tau_a \ll T_2^* \\ \dfrac{\Gamma_a}{\Gamma_n} = \dfrac{\pi T_2^*}{\tau_a}\,, & T_2^* \ll \tau_a \end{cases}. \tag{2.10}$$

The r.m.s. tipping angle $\sqrt{\langle \xi^2 \rangle}$ is limited by the interplay between ALP decoherence and spin relaxation. Modeling the ALP decoherence as a random phase that resets on the time scale τ_a, the tipping angle executes a two-dimensional random walk in the rotating frame with step size $\Omega_a \tau_a$ [32]. Combining this with the three relevant relaxation scales,

$$\sqrt{\langle \xi^2 \rangle} = \begin{cases} \Omega_a \sqrt{\tau_a T_2^*}, & \tau_a \ll T_2^* \\ \Omega_a \dfrac{\tau_a}{\sqrt{\pi}}, & T_2^* \ll \tau_a \ll T_2 \\ \Omega_a T_2, & T_2 \ll \tau_a \end{cases}. \tag{2.11}$$

These cases combine into the r.m.s. transverse magnetization (for dwell time $T \gg \tau_a, T_2^*, T_2$):

$$\sqrt{\langle M_1^2 \rangle} \approx \frac{M_0 \Omega_a}{\sqrt{2}} \begin{cases} \sqrt{\tau_a T_2^*}\,, & \tau_a \ll T_2^* \\ \dfrac{T_2^*}{\sqrt{\pi}}\,, & T_2^* \ll \tau_a \ll T_2 \\ \dfrac{T_2^* T_2}{\sqrt{\pi}\,\tau_a}\,, & T_2 \ll \tau_a \end{cases}. \tag{2.12}$$

In the third case, when $T_2 \ll \tau_a$ and the dwell time $T \gg \tau_a$, the signal amplitude decreases because the ALP field cannot drive the spins coherently for longer than T_2, while off-resonance spins (those not within the narrow ALP linewidth) contribute negligibly. Yuzhe: Add numerical simulation results here; reference Appendix A for details.

2.1.5 Summary of relaxation and coherence effects

Depending on the relative magnitudes of T_2, T_2^*, and τ_a, four regimes are identified [32]:

(1) $\tau_a \ll T_2^*$;

(2) $T_2^* \ll \tau_a \ll T_2$;

(3) $T_2^* \ll T_2 \ll \tau_a$;

(4) $T_2^* = T_2 \ll \tau_a$.

These regimes are illustrated in Figure 2.1. The key parameters for each regime are summarized in Table 2.1, where ν_{start} and ν_{end} are the lower and upper limits of the target frequency band.

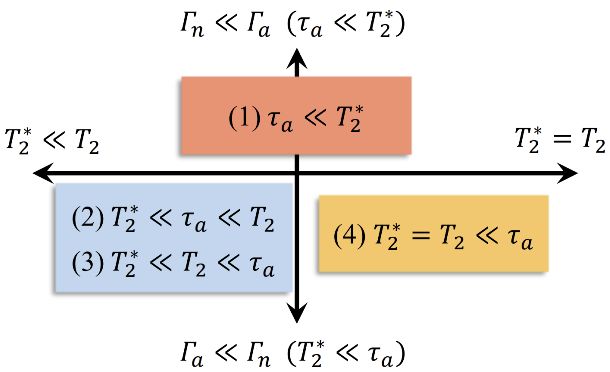

Figure 2.1: Regimes of the spin-precession experiment based on the relative magnitudes of T_2, T_2^*, and τ_a. Reproduced from Ref. [32].

Yuzhe: check if optimal scan strategy is mentioned in this subsection

2.1.6 Periodic modulations

Yuzhe: Use Gramolin paper and mention briefly about Periodic modulation. mention the modulation that terrestrial detectors can feel.

Table 2.1: Summary of parameters for the four regimes. The normalized linewidth $\delta_\nu = \tau_a/(\pi Q_a T_2^*)$ is assumed constant in each regime. The optimal dwell time is derived in Appendix C.

	Regime	Optimal dwell time T	Signal linewidth Γ	Spectral factor u_n	r.m.s. ti
(1)	$\tau_a \ll T_2^*$	$\dfrac{T_{\text{tot}}}{Q_a \ln(\nu_{\text{end}}/\nu_{\text{start}})}$	$\dfrac{2}{T_2^*}$	1	
(2)	$T_2^* \ll \tau_a \ll T_2$	$\dfrac{T_{\text{tot}}\,\tau_a}{\pi Q_a T_2^* \ln(\nu_{\text{end}}/\nu_{\text{start}})}$	$\dfrac{2\pi}{\tau_a}$	$\dfrac{\pi T_2^*}{\tau_a}$	
(3)	$T_2^* \ll T_2 \ll \tau_a$	$\dfrac{T_{\text{tot}}\,\tau_a}{\pi Q_a T_2^* \ln(\nu_{\text{end}}/\nu_{\text{start}})}$	$\dfrac{2\pi}{\tau_a}$	$\dfrac{\pi T_2^*}{\tau_a}$	
(4)	$T_2^* = T_2 \ll \tau_a$	$\dfrac{T_{\text{tot}}\,\tau_a}{\pi Q_a T_2 \ln(\nu_{\text{end}}/\nu_{\text{start}})}$	$\dfrac{2\pi}{\tau_a}$	$\dfrac{\pi T_2}{\tau_a}$	

2.1.7 Numerical simulation

The analytical estimates above are valid in the limit $T \gg \tau_a$, T_2^*, T_2. At lower Compton frequencies, the ALP coherence time can become comparable to or longer than experimentally accessible dwell times, and the simple scaling laws no longer apply. Numerical simulations using the `axionbloch` package (Appendix A) are used to validate the analytical estimates and to explore intermediate parameter regimes. Numerical simulation is then the tool of choice.

The package models spin-1/2 ensembles through the Bloch equations (1.4)–(1.6) derived in Section 1.1.2. The total effective field is $\mathbf{B}_{\text{eff}} = \mathbf{B}_0 + \mathbf{B}_a(t)$, combining the laboratory bias field and the axion-induced pseudomagnetic field. The equations are solved in the rotating coordinate frame (RCF), which removes the fast Larmor precession and exposes the slow axion-induced dynamics. The effective transverse relaxation time T_2^* (which includes inhomogeneous broadening) is simulated by sampling the spread of bias-field values from a `Magnet` instance and integrating the Bloch equations for each sampled field independently.

The time evolution of $\mathbf{M}$ is obtained by numerically integrating Bloch equations

[Equations (1.4–1.6)] with the fourth-order Runge–Kutta (RK4) method. Given a time step Δt and initial magnetization $\mathbf{M}^n$, the update rule is

$$\mathbf{M}^{n+1} = \mathbf{M}^n + \frac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right), \tag{2.13}$$

where $\mathbf{k}_1 = d\mathbf{M}/dt|_{\mathbf{M}^n}$ and subsequent $\mathbf{k}_i$ are evaluated at intermediate half-steps following the standard RK4 prescription. The local truncation error is $O(\Delta t^5)$ and the global accumulated error is $O(\Delta t^4)$. The time step is set to be at least one order of magnitude smaller than the shortest characteristic timescale in the simulation (e.g., T_1, T_2, or $1/\nu_a$).

The RK4 integration is implemented in a compiled C++ backend and exposed to Python through the `pybind11` interface, achieving runtimes of ~ 10 s for $\approx 4\times10^9$ integration steps on a modern desktop CPU. The details of the implementations are included in the appendix. This appendix describes `axionbloch`, an open-source Python package that simulates spin dynamics under both magnetic and axion-induced pseudomagnetic fields.

2.2 Thermalized Milky Way ALP

Yuzhe: ideal gas treatment: discuss thermalized halo and velocity distribution.

The r.m.s. Rabi frequency Ω_a can be computed based on Equations (1.9), (1.11), and (1.10) [11]:

$$\Omega_a = \frac{1}{2\hbar} g_{\mathrm{aNN}} a_0 m_a c v_\perp \approx \frac{1}{2} g_{\mathrm{aNN}} \sqrt{2\hbar c \rho_{DM}} v_\perp \,, \tag{2.14}$$

Here we assume that the ALP-field gradient comes exclusively from the relative motion of the field and the detector.[2] If we take $v_\perp = 220\,\mathrm{km\,s^{-1}}$, we have $\Omega_a/2\pi = g_{\mathrm{aNN}} \times 0.2\,\mathrm{GeV}\cdot\mathrm{Hz}$. For an experiment searching for $g_{\mathrm{aNN}} \leq 10^{-10}\,\mathrm{GeV^{-1}}$, the Rabi period is $2\pi/\Omega_a \geq 1000\,\mathrm{yr}$; the ALP-field gradient is therefore a weak drive on the spins.

2.2.1 Coherence time

For a Maxwell-Boltzmann velocity distribution with average speed $v_a \approx 220\,\mathrm{km\,s^{-1}}$ [15], the ALP field spectral linewidth (FWHM) observed by a detector on Earth is

$$\Gamma_a = \frac{\nu_a}{Q_a} \,, \tag{2.15}$$

where $Q_a \equiv (c/v_a)^2 \approx 1.8 \times 10^6$ is the ALP quality factor. The ALP coherence time τ_a, defined by $\Gamma_a = 2\pi/\tau_a$, is therefore

$$\tau_a = \frac{2\pi Q_a}{\nu_a} \,. \tag{2.16}$$

Within one coherence time the superposition Equation (2.5) is well approximated by the single coherent mode Equation (1.9). On longer time scales the random phases between modes cause the amplitude and phase of the effective oscillation to drift, so the field must be treated as a stochastic process [11]. For Compton frequencies above 1 kHz, $\tau_a \lesssim 1000\,\mathrm{s}$; the interplay between τ_a and the NMR relaxation times determines the optimal measurement strategy in Chapter 4.

2 This is not necessarily the case, for instance, for ALPs gravitationally bound in stationary states like axion stars or "axion atoms" [**Lam2017_PhysRevD**, **Liang_Zhitnitsky2019_GravitationallyPhysRevD**]. We do not consider such regimes here.

2.2.2 Periodic modulations

The ALP-field gradient driving spin precession depends on the direction of the relative velocity between the Earth and the local dark-matter halo. As the Earth rotates, this direction sweeps through space with a period of one sidereal day. An ALP signal therefore exhibits a daily modulation of its amplitude and phase [18].

In addition to daily rotation, the Earth's orbital velocity around the Sun modulates the relative ALP-detector velocity with a period of one year. Both the daily and annual modulations have precisely predictable periods and phases, making them useful consistency checks for any claimed dark-matter-candidate signal [18].

2.2.3 Numerical simulation

Yuzhe: include several important examples and give a summary of signal dependence.

2.3 Earth-bound ALPs

The galactic dark-matter halo provides the primary ALP signal target discussed in Chapter 4. However, a distinct and potentially enhanced source arises from ALPs that are gravitationally bound to the Earth. Such Earth-bound states are analogous to atomic bound states—the Earth's gravitational potential plays the role of the Coulomb well and the ALP plays the role of the electron—and can support a discrete spectrum of spatial eigenstates. This section derives the properties of these bound states, estimates the resulting spin-precession signal, and discusses the characteristic daily modulation that distinguishes this signal from the galactic background.

2.3.1 Trapping and accumulation of dark matter

For Earth-bound states to be populated, ALPs from the galactic halo must lose enough kinetic energy to fall below the local escape velocity ($v_{\mathrm{esc}} \approx 11.2\,\mathrm{km\,s^{-1}}$ for Earth). Several mechanisms have been proposed Yuzhe: cite: Banerjee et al. 2019 arXiv:1902.07720; OHare 2025 fine-grained; other relevant papers:

Gravitational three-body scattering A galactic ALP passing near the Earth can scatter off a third body (e.g. the Sun or Moon), losing kinetic energy and becoming trapped in a bound orbit. The rate depends on the local density and the ALP-body cross section.

Self-interaction or wave-packet spreading Yuzhe: placeholder: if ALPs have self-interactions or if wave-packet decoherence provides an effective dissipation, this may also populate bound states.

Resonant capture Near specific orbital resonances the exchange of energy between the ALP and the Earth-Moon or Earth-Sun system may be efficient.

The total mass M_{bound} of ALPs trapped in Earth-bound states is strongly constrained by astrophysical and geophysical observations Yuzhe: leave the plot here. cite and expand: relevant constraints on exotic mass inside Earth from geophysics / seismology / gravitational measurements. An upper bound comes from the requirement that the Earth's measured gravitational multipole moments not be distorted beyond current geodetic precision. Yuzhe: Add quantitative upper limit on M_{bound} once reference is identified. Yuzhe: We note that.... skip the mechanism of capturing...

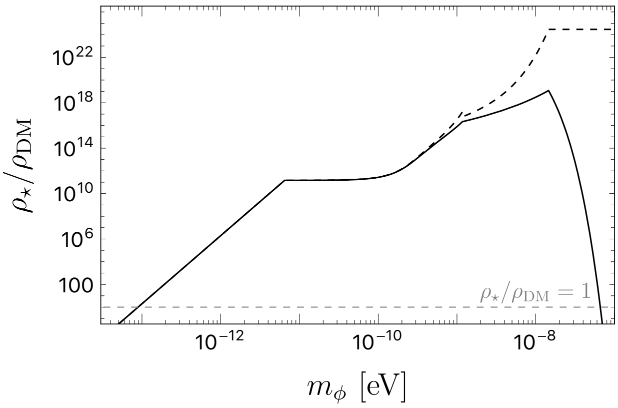

Figure 2.2: Density Distribution of ground state Earth-bound ALP halo. The solid line indicates the density at the Earth surface, whereas the dashed line represents the density at the center of the halo. The density value $\rho_\star$ is derived from the maximum Earth ALP halo mass permitted by gravitational constraints, as assessed in [6, 7].

2.3.2 Gravitational potential and the quantum picture

To simplify the calculation, Earth is considered as a spherical object. The density of Earth varys over radii. From geophysical measurements on earthquake waves, Preliminary reference Earth model (PREM)[14] was constructed, giving the Earth density profile as shown in Figure 2.3.

The gravitational potential of Earth V_g/m per unit mass can be found by

$$V_g(r)/m = -\frac{GM_{\rm enclosed}}{r}\,, \tag{2.17}$$

Yuzhe: check if we are wrong by a constant or so. where r is the distance measured from the Earth's center, G is Newton's constant, and $M_{\rm enclosed}$ is the Earth's mass enclosed by r. Here we take the infinity as the reference point and have potential $V_g(\infty)/m = 0\,\mathrm{J\,kg^{-1}}$. The potential over radii can be calculated numerically based on PREM (Figure 2.3). The Earth escape velocity can therefore be found by

$$v_{\rm escape} = \sqrt{\frac{2GM_\oplus}{R_\oplus}}\,, \tag{2.18}$$

where $M_\oplus = 6\times 10^{24}\,\mathrm{kg}$ and $R_\oplus = 6378.1\,\mathrm{km}$ are the mass and average radius of Earth, respectively. By plugging the values, the Earth escape velocity is found

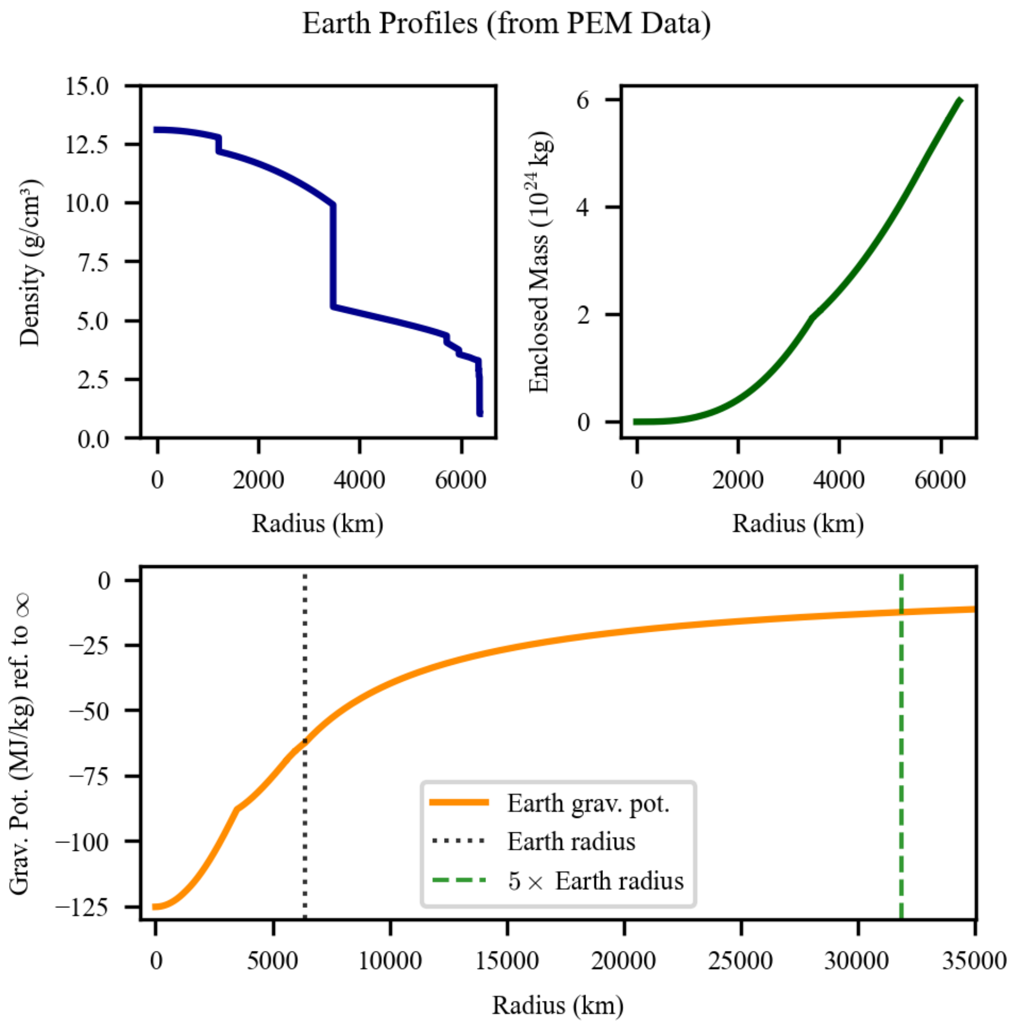

Figure 2.3: to be modified later.

to be 11.2 km/s. Since the escape speed is much smaller than the speed of light, non-relativistic limit is assumed for the discussion.

For ultralight dark matter particles bound by Earth, they can no longer be treated as classic objects due to their large de Broglie wavelengths. A rough estimation of the wavelength is made as follows. The ALP masses that CASPEr-G can probe range from $\sim$ 4 peV to 2.5 μeV. Consider ALPs trapped by the earth potential with velocities roughly smaller than the escape veclocity. The de Broglie wavelength as a function of particle mass $\lambda_{\mathrm{dB}}(m_a, v_a) = \frac{h}{m_a v_a}$ gives

- $\lambda_{\mathrm{dB}}(4\,\mathrm{peV}, 11.2\,\mathrm{km/s}) = 8 \times 10^9\,\mathrm{m} = 1260\,\mathrm{R}_\oplus$ and
- $\lambda_{\mathrm{dB}}(2.5\,\mu\mathrm{eV}, 11.2\,\mathrm{km/s}) = 3\,\mathrm{mm}$.

Therefore, in the scan range of CASPEr, the de Broglie wavelength could be comparable to or much larger than the earth radius. In later chapters, a measurement probing MHz ALPs is presented. This corresponds to a de Broglie wavelength $\sim 1\,\mathrm{R}_\oplus$. To prepare for the analysis, we focus on this scenario and take the wave nature into account. Yuzhe: mention Na here.

2.3.3 Bound-state wavefunctions

The ALP field consists of superpositions of eigenstates

$$a_0(\mathbf{r}, t) = \sqrt{\frac{N_a \hbar^3 c}{2m_a}} \left(\Psi(\mathbf{r}, t) + \Psi^*(\mathbf{r}, t) \right) , \tag{2.19}$$

where

$$\Psi(\mathbf{r}, t) = e^{-i m_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-i E_n t/\hbar} \psi_{nlm}(\mathbf{r}) . \tag{2.20}$$

Here N_a is the number of ALP particles, $\Psi(\mathbf{r}, t)$ is the wavefunction, ψ_{nlm} and E_n are the eigen-wavefunction and the eigen-energy of the eigenstate (n, l, m), c_{nlm} are the expansion coefficients determined by the population of eigenstates, n is the principal quantum number, l is the angular momentum quantum number, and m is the magnetic quantum number. The fast oscillation at ALP Compton frequency is indicated by $e^{-i m_a c^2 t/\hbar}$, where the small variation of oscillation frequency is indicated by $e^{-i E_n t/\hbar}$. Since ALP is a real scalar field, Equation 2.19 can be rewritten as

$$a_0(\mathbf{r}, t) = \sqrt{\frac{2 N_a \hbar^3 c}{m_a}} \Re \left[\Psi(\mathbf{r}, t) \right] , \tag{2.21}$$

where $\Re$ indicates the real part. The ALP dark matter energy density is $\rho^E_{DM} = m_a c^2 N_a \langle|\Psi|^2\rangle_t$, where $N_a\langle|\Psi|^2\rangle_t$ is the time-averaged ALP number density. With ρ^E_{DM}, the ALP field can be expressed as

$$a_0(\mathbf{r},t) = \sqrt{\frac{2\hbar^3\rho^E_{DM}}{m_a^2 c}}\frac{\Re[\Psi(\mathbf{r},t)]}{\langle|\Psi(\mathbf{r},t)|^2\rangle_t^{1/2}} \tag{2.22}$$

$$= a_0\frac{\Re[\Psi(\mathbf{r},t)]}{\langle|\Psi(\mathbf{r},t)|^2\rangle_t^{1/2}}\,. \tag{2.23}$$

Here a_0 is the ALP field amplitude, of which the dimension is energy. From astronomical observations, the local dark matter energy density can be estimated. Yuzhe: TODO: metion Na Therefore, the ALP field amplitude can be obtained by Equation 2.23 once the wavefunction is derived.

In the non-relativistic limit, the eigenstates can be found by the time-independent Schrödinger equation (TISE)

$$\left[-\frac{\hbar^2}{2m_a}\nabla^2 + V_g(r)\right]\psi_{nlm}(\mathbf{r}) = E_n\,\psi_{nlm}(\mathbf{r})\,. \tag{2.24}$$

In analogy to the hydrogen atom, the ALP-Earth system is a gravitational atom Yuzhe: cite: relevant paper on gravitational atom, e.g. Arvanitaki et al. and more.... Yuzhe: what about quantum number m? zero?

Considering the spherical symmetry of the Earth gravitational potential, spherical coordinate system (r,θ,φ) is adopted, where r is the radial distance from the origin, θ is the polar angle, and φ is the azimuthal angle. In spherical coordinates,

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\,, \tag{2.25}$$

so that TISE can be rewritten as

$$\left[-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}\right) + V_g(r)\right]\psi_{nlm}(r,\ \theta,\varphi) = E_n\psi_{nlm}(r,\ \theta,\varphi)\,, \tag{2.26}$$

where $\hat{L}$ is the angular momentum operator. The eigen-function can be separated as

$$\psi_{nlm}(\mathbf{r}) = R_{nl}(r)\,Y_l^m(\hat{\mathbf{r}})\,, \tag{2.27}$$

where R_{nl} is the radial wavefunction and Y_l^m is the spherical harmonic. The spherical harmonics satisfies

$$\hat{L}^2 Y(\theta,\ \varphi) = l(l+1)\hbar^2 Y(\theta,\varphi)\,. \tag{2.28}$$

Therefore, to derive the radial wavefunction of eigen-state with angular quantum number l, TISE is transformed to

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R_{nl}(r)}{\partial r}\right) + \left(\frac{l(l+1)\hbar^2}{2mr^2} + V_g\right) R_{nl}(r) = E_n R_{nl}(r)\,. \tag{2.29}$$

Since the earth density is not uniform, it is difficult to derive the radial wavefunction analytically. Hence, numerical method is adopted. In a discretized space, a matrix for the Hamiltonian in Equation (2.29) is constructed. The eigen-values (eigen-energies) and eigen-vectors (eigen-states) can be derived using Python package `SciPy`. This method is implemented in the Python class `EarthBoundAxionHalo` of `axionbloch`.

Note that normalization is necessary for deriving the right magnitude of the eigen-functions. The wavefunctions are normalized so that the total probability integrates to unity,

$$\int |\psi_{nlm}(\mathbf{r})|^2\, d^3r = \int_0^\infty |R_{nl}(r)|^2\, r^2\, dr \int |Y_l^m(\hat{\mathbf{r}})|^2\, d\Omega = 1\,. \tag{2.30}$$

Here Ω is solid angle coordinate. Examples of normalized wavefunctions are illustrated in Figure 2.4. Yuzhe: eigen states plotted here:

2.3.4 ALP field gradient

The ALP field gradient at the detector can be derived from the spatial gradient of the wavefunction ψ_{nlm} evaluated at the detector position. The gradient operator in spherical coordinates is

$$\boldsymbol{\nabla} = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\boldsymbol{\varphi}}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\,. \tag{2.31}$$

The gradient of the ALP field can therefore be inferred from Equation (2.21) and (2.20) as

$$\boldsymbol{\nabla} a(\mathbf{r},t) = \sqrt{\frac{N_a\hbar^3 c}{2m_a}}\,\Re(\boldsymbol{\nabla}\Psi(\mathbf{r},t)) \tag{2.32}$$

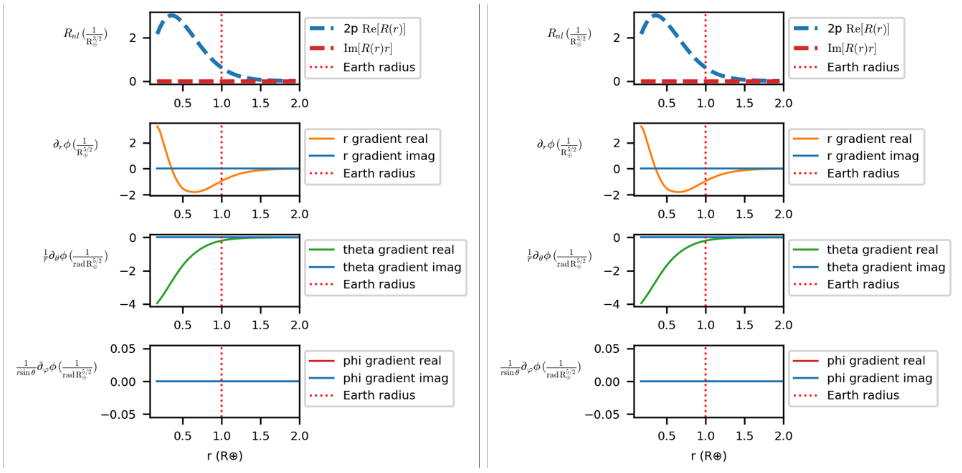

Figure 2.4: radial wavefunctions of 1s 2p... states. include higher states and mention that we are only interested in low energy states.

$$= \sqrt{\frac{N_a \hbar^3 c}{2m_a}} \Re \left(e^{-im_a c^2 t/\hbar} \sum_{nlm} c_{nlm} e^{-iE_n t/\hbar} \nabla \psi_{nlm}(\mathbf{r}) \right), \tag{2.33}$$

where

$$\nabla \psi_{nlm}(\mathbf{r}) = \hat{\mathbf{r}} \frac{\partial R_{nl}}{\partial r} Y_l^m + \hat{\boldsymbol{\theta}} \frac{R_{nl}}{r} \frac{\partial Y_l^m}{\partial \theta} + \hat{\boldsymbol{\varphi}} \frac{R_{nl}}{r \sin\theta} \frac{\partial Y_l^m}{\partial \varphi}. \tag{2.34}$$

If multiple modes are populated, the total gradient is a superposition of the gradients from each mode, weights of which correspond to the population distribution, as can be seen in Equation (2.33).

contributed by the wavefunction ψ_{nlm}

The gradient magnitude depends on the eigen-state (n, l, m) and the detector location. The characteristics of the gradient are addressed as follows:

- For the eigen-states with zero angular momentum (s state), the wavefunction is spherically symmetric, so the gradient is purely radial and its magnitude is proportional to the radial wavefunction gradientYuzhe: is this paragraph right??.
- When angular momentum is non-zero ($l >= 1$), the gradient has non-radial components may vary over different locations, depending on the angular structure of the wavefunction.
- The choice of the quantization axis is critical for considering $l > 0$ states. Since the solar gravity breaks the rotational symmetry, the quantization axis

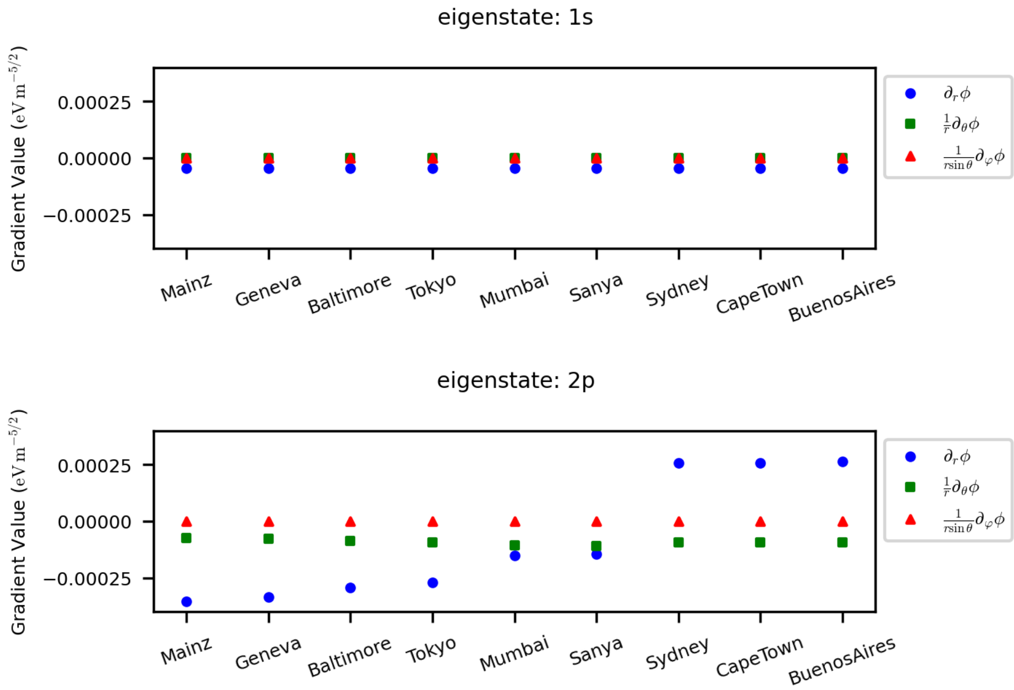

Figure 2.5: gradients of 1s 2p... states. Three rows would be enogh.

(z-axis) is chosen along the direction from the Earth's center to the Sun, which does not align with the Earth's rotational axis. This means that the gradient at the detector location will have a daily modulation as the Earth rotates, which is discussed in Section 2.3.6.

- **Azimuthal dependence** For modes with $m \neq 0$, the gradient has an azimuthal dependence. However, for the brevity of this discussion, only $m = 0$ is considered in this work.

The values of the gradients are computed numerically from the eigen-states and are used to estimate the expected signal strength in Chapter 4. Yuzhe: mention the time for computing the gradients. Examples of the gradient magnitude for selected modes are illustrated in Figure 2.4.

Yuzhe: Add numerical estimate for $\Omega_a^{\mathrm{Earth}}$ as a function of m_a and M_{bound}. Add comparison with the galactic-halo Rabi frequency Ω_a (Equation (1.14)). A rough estimation of the Rabi frequency $\Omega_a^{\oplus}$ of the pseudomagnetic field can be made using Equation (1.14) and escape speed v_{esc} as the typical velocity of the Earth-bound ALP:

$$\Omega_a^{\oplus}/(2\pi) \approx \frac{g_{\mathrm{aNN}}}{10^{-9}\,\mathrm{GeV}^{-1}} \times 6\,\mathrm{mHz}\,. \tag{2.35}$$

A more accurate estimation can be made using the numerical values of the wavefunction gradients at the detector location. With Equations (1.13, 2.33, 2.34), it can be estimated as

$$\Omega_a^{\oplus} = c\, g_{\mathrm{aNN}} \sqrt{\frac{N_a \hbar^3 c}{2m_a}}\, |(\boldsymbol{\nabla}\psi_{nlm})_{\perp}|\,, \tag{2.36}$$

where $|(\boldsymbol{\nabla}\psi_{nlm})_{\perp}|$ is the magnitude of the wavefunction gradient component perpendicular to the bias field $\mathbf{B}_0$ (i.e., the $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\varphi}}$ terms from Equation (2.34)). Using the numerical values of the wavefunction gradients, the value can be obtained as

$$\Omega_a^{\oplus,\theta}/(2\pi) \approx \frac{g_{\mathrm{aNN}}}{10^{-9}\,\mathrm{GeV}^{-1}} \times 3\,\mathrm{mHz}\,. \tag{2.37}$$

Yuzhe: what about r gradient? TODO: add it These two estimations yield consistent results within an order of magnitude. The calculations can be found at Yuzhe: add reference to the script used for the calculation. Note that due to the magnitude of the Rabi frequency can vary by 100 % due to the time-changing alignment between the ALP gradient and the sensitive axis of detector.

Such Rabi frequency is ~7 orders of magnitude larger than the expected Rabi frequency from the galactic halo ALP, which is $\Omega_a/(2\pi) = \frac{g_{\mathrm{aNN}}}{10^{-9}\,\mathrm{GeV}^{-1}} \times 2 \times 10^{-10}$ Hz, as estimated after Equation (1.14)Yuzhe: TODO: I may have to choose a different equation after finishing theory section. If indeed such amount of axionlike dark matter can be accumulated in the Earth gravitational potential, it would produce a much stronger signal than the galactic halo ALP, making it an target that CASPEr-G is sensitive to.

2.3.5 Population and coherence

Unlike the galactic halo signal, which has a broad Maxwell-Boltzmann spectrum, Earth-bound ALPs occupy discrete energy levels E_n. Each level n contributes an ALP oscillation at a slightly different Compton frequency

$$\nu_n = \nu_a + \frac{|E_n|}{h} = \nu_a \left(1 + \frac{\alpha_g^2}{2n^2}\right), \tag{2.38}$$

where $\nu_a = m_a c^2/h$ is the rest-mass Compton frequency. The level spacing between adjacent states n and $n+1$ is

$$\Delta\nu_n = \frac{m_a c^2\,\alpha_g^2}{h}\left(\frac{1}{2n^2} - \frac{1}{2(n+1)^2}\right) \approx \frac{m_a c^2\,\alpha_g^2}{h\,n^3}\,. \tag{2.39}$$

Yuzhe: Add numerical value of $\Delta\nu_n$ for $n = 1$ and the masses of interest.

If multiple modes are populated simultaneously, their superposition produces a beating pattern with a period $\tau_{\text{beat}} \approx 1/\Delta\nu_n$. This plays the role of a coherence time for the multi-mode Earth-bound field: the signal amplitude fluctuates on the time scale τ_{beat}, analogous to the galactic-halo coherence time $\tau_a = Q_a/\nu_a$ but with a discrete rather than continuous spectrum.

Numerical solutions of Equation (2.24) are needed to determine the wavefunctions and gradients for non-spherical or higher-l modes and to account for the finite density profile of the Earth Yuzhe: placeholder: add reference to numerical solutions and/or to the script used to compute them; planned figure showing probability density and gradient magnitude for selected (n, l, m) modes.

2.3.6 Daily modulations

In the resonance scheme, CASPEr-G is only sensitive to the component of $\boldsymbol{\nabla} a$ perpendicular to the bias field $\mathbf{B}_0$. Since the bias field is always veritcal with repect to the ground, the setup is sensitive to gradient along $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\varphi}}$ directions, but not to the gradient along $\hat{\mathbf{r}}$ direction.

assuming the ALP halo is stationary while Earth is rotating at period of a day, daily modulation of the ALP signal amplitude is expected. However, the numerical

The Earth-bound ALP wavefunction ψ_{nlm} is defined in the Earth-centered inertial frame, with the quantization axis aligned with the Earth's rotation axis (or with some fixed direction in space, depending on the angular momentum quantum numbers l, m). As the Earth rotates, the laboratory co-rotates, so the orientation of the gradient $\boldsymbol{\nabla} a$ relative to the bias field $\mathbf{B}_0$ changes with a period of one sidereal day.

The component of $\boldsymbol{\nabla} a$ perpendicular to $\mathbf{B}_0$ (which drives nuclear spin precession, cf. Equation (1.14)) therefore exhibits a daily modulation. For a bias field pointed along the local vertical, and a bound state with quantum numbers (n, l, m), the perpendicular gradient component traces a specific daily pattern determined by the spherical harmonic Y_l^m evaluated at the detector's co-latitude Yuzhe: add explicit formula for the modulation amplitude as a function of co-latitude and (l, m).

This daily modulation is a powerful discriminant: its period (one sidereal day) and phase are fixed by Earth's rotation and the quantum numbers of the bound state, providing a strong consistency check that is qualitatively different from both the galactic-halo signal (which has a different daily modulation driven by the velocity of the halo wind, cf. Section 2.3.6) and instrumental artifacts. Yuzhe: Add figure: predicted daily modulation envelope for the ground state at a mid-latitude detector, for one or two representative masses.

3 Experiment

The apparatus description is adapted from Ref. [28]. Sections 3.7–3.9 are adapted from Ref. [32].

The CASPEr-Gradient-Low-Field apparatus converts a nuclear-spin signal into a detectable electrical output through three functional blocks: the magnet system sets the operating bias field and thereby the Larmor frequency of the sample; the sample and pickup coil generate and capture the magnetic response; and the SQUID and lock-in amplifier chain convert, amplify, and record the signal. This chapter describes each block, summarizes the key parameters, and then develops the strategy used to scan an ALP-mass range efficiently.

3.1 Magnet system

The apparatus is built around a flow-through liquid-helium cryostat housing a superconducting solenoid that can generate a bias field B_0 up to 0.1 T. The DM mass range accessible to the experiment depends on the gyromagnetic ratio of the sample and on B_0; for proton spins, the covered Larmor frequency range is approximately 1 kHz to 4.3 MHz.

Eight superconducting shim coils surrounding the solenoid are used to reduce the field inhomogeneity to less than 2 ppm over a spherical volume with radius 4 mm. In the measurements reported here, an inhomogeneity of approximately 10 ppm was achieved after tuning the shim fields using a Bayesian-optimization algorithm [27]. The larger-than-design inhomogeneity results from using a longer sample (24 mm height) and from the insertion of an additional superconducting pickup coil.

Helmholtz coils wound outside the cryostat provide a controllable offset on B_0, enabling fine-tuning of the Larmor frequency without ramping the main solenoid. This arrangement is used to step the Larmor frequency in 6 Hz increments during the scan.

3.1.1 Cryostat and superconducting magnet

The cryostat housing the superconducting magnet is placed inside a Faraday room, shielded from external electric fields. It consists of an outer shell, reservoirs for liquid nitrogen (LN_2) and liquid helium (LHe), and vacuum insulation layers in between. Liquid nitrogen cools the cryostat and reduces the evaporation of liquid helium, while the superconducting devices and niobium shield are submerged in the LHe bath at 4.2 K. When filled, the cryostat can maintain a superconducting environment for approximately one week. The niobium shield repels external magnetic fields, isolating internal devices from external magnetic noise.

The magnet system consists of the main solenoid coil and eight shim coils (named Z1, Z2, X, Y, XY, XZ, YZ, and X^2-Y^2), each capable of generating magnetic field components for field homogenization. The shim coil channel names indicate the geometry of the magnetic field they generate, enabling systematic correction of field inhomogeneities.

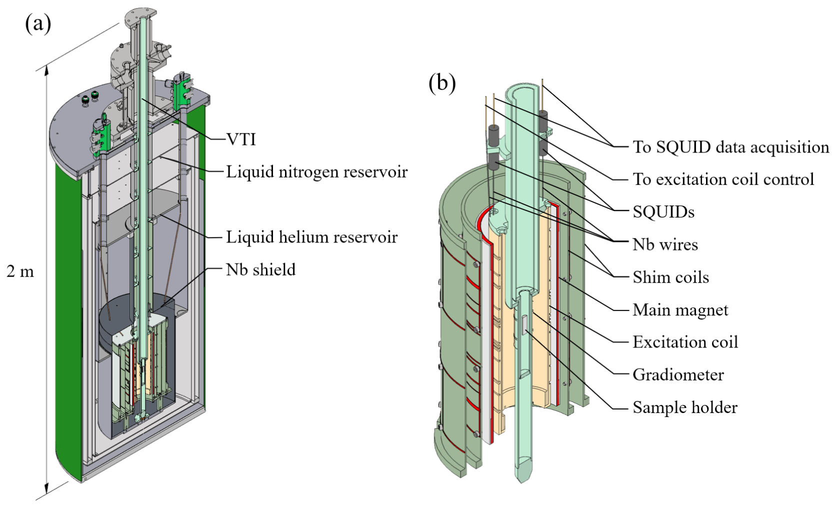

Figure 3.1: (a) Cryostat for maintaining superconducting temperature. LHe bath provides 4.2 K environment for superconducting apparatus. (b) Superconducting apparatus including main coil, shim and excitation coils, SQUIDs with niobium capsule, niobium wires for connections, and gradiometer made of niobium wires.

3.1.2 Excitation generation

To perform pulsed-NMR schemes, an excitation coil made of niobium wires is positioned near the sample to avoid Johnson–Nyquist noise from thermal agitation in normal-conducting materials. The excitation coil provides B_x, B_y, B_z, and dB_z/dz channels, with the first three capable of producing homogeneous fields along the respective axes and the last generating a linear gradient along the z-axis.

Radiofrequency pulses are generated by a function generator and amplified by an rf amplifier. A tuning circuit composed of capacitors and inductors is designed for impedance matching between the function generator and the excitation coil across different frequencies.

3.2 Sample

Liquid methanol (CH_3OH) was chosen as the sample because of its high proton density (99 mmol/cm^3) and low freezing point (175.6 K), which allows it to remain liquid at the operating temperature of approximately 180 K. With four protons per molecule, the 1.2 cm^3 cylindrical sample (radius 4 mm, height 24 mm) contains 7.2×10^{22} protons.

A variable-temperature insert (VTI) positions the sample in the homogeneous region of the bias field and thermally isolates it from the liquid-helium bath. The sample temperature is regulated by flowing warm nitrogen gas through a spiral surrounding the sample holder. Temperature stability is monitored by repeated pulsed-NMR measurements: a constant NMR linewidth indicates a stable temperature.

At thermal equilibrium in the bias field B_0, the proton spins acquire a net longitudinal magnetization

$$M_0 = \frac{n\,\gamma^2\hbar^2\, I(I+1)\, B_0}{3\,k_B\,T}\,, \tag{3.1}$$

where n is the proton number density, $\gamma = 2\pi \times 42.577\,\mathrm{MHz/T}$ is the proton gyromagnetic ratio, $I = 1/2$, k_B is the Boltzmann constant, and T is the sample temperature. At $B_0 = 31.67\,\mathrm{mT}$ and $T \approx 180\,\mathrm{K}$, the thermal polarization is $\approx 1.8 \times 10^{-7}$ and $M_0 \approx 1.86 \times 10^{-10}\,\mathrm{T}/\mu_0$.

3.3 Pickup coil and shielding

A niobium (Nb) pickup coil mounted on the outside of the VTI in the liquid-helium bath detects the transverse magnetization of the sample. The coil consists of three saddle-pair coils in a gradiometer configuration: the area of the central coil is twice that of the two outer coils, and the coils are wound in opposing senses so that far-field flux contributions cancel. The filling factor—the fraction of the coil volume occupied by the sample—is approximately 8 %.

A superconducting Nb shield built into the cryostat attenuates low-frequency magnetic noise. Additional shielding is provided by a μ-metal shield outside the cryogenic reservoir and by a Faraday cage enclosing the entire apparatus, which suppresses electromagnetic interference (EMI). The pickup coil geometry and pulsed-NMR scheme are illustrated in Figure 3.3.

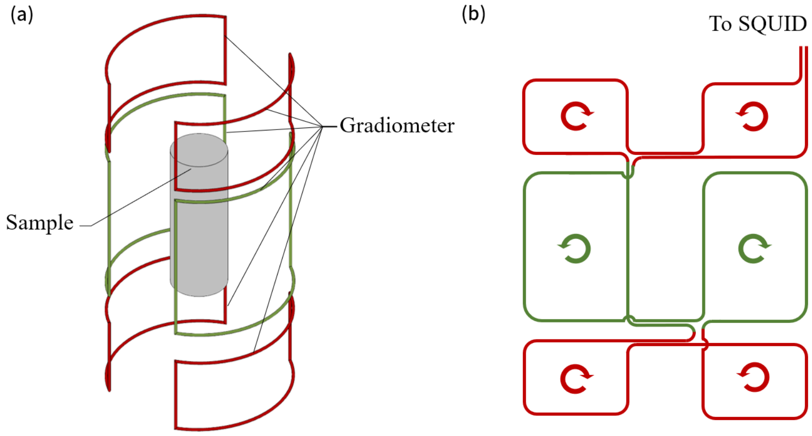

Figure 3.2: (a) Simplified schematic of gradiometer and sample positioned at the center. (b) Details of gradiometer loop connection showing the three-coil saddle-pair configuration with opposing windings for noise rejection.

3.4 Detection chain

3.4.1 SQUID detector and flux-locked-loop operation

Magnetic flux in the pickup coil is inductively coupled into a direct-current superconducting quantum interference device (dc-SQUID) operated in flux-locked-loop (FLL) mode. The SQUID converts magnetic flux into electrical signals through the combination of flux quantization and Josephson tunneling effects, offering exceptional sensitivity for weak magnetic signals.

The relationship between the magnetic flux coupled to the SQUID and the feedback voltage is given by equation (3.2), where the conversion factor depends on the coupling strengths and feedback parameters summarized in Table 3.1. The SQUID circuit is enclosed in a niobium capsule for shielding and protection from external disturbances.

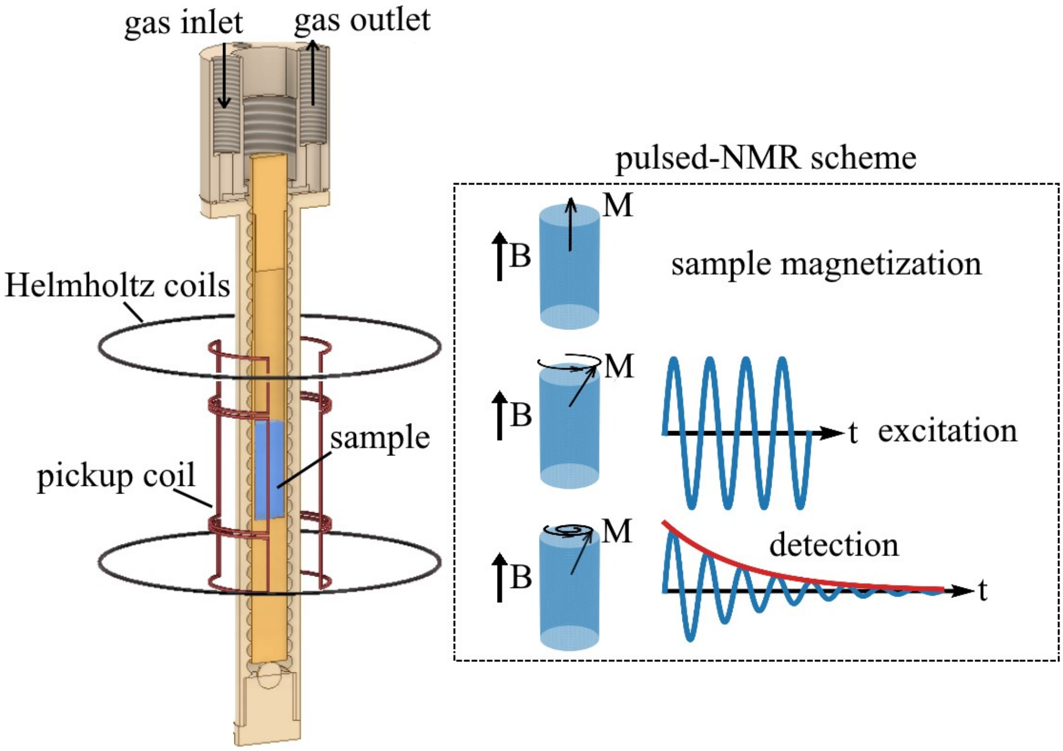

Figure 3.3: Left: the methanol sample (blue) sits at the center of the three-coil Nb saddle-pair pickup (red). Helmholtz coils (black) fine-tune the bias field. Right: pulsed-NMR scheme used for sample characterization. The net magnetization **M** is tipped into the transverse plane by a $\pi/2$ pulse and precesses, generating a signal in the pickup and SQUID. Reproduced from Ref. [28].

3.4.2 Signal chain and processing

Magnetic flux in the pickup coil is inductively coupled into a direct-current superconducting quantum interference device (dc-SQUID) operated in flux-locked-loop (FLL) mode. The properties of the SQUID sensor are listed in Table 3.1. The feedback voltage V_f is related to the magnetic flux in the pickup coil Φ_pick by

$$V_\mathrm{f} = \frac{R_\mathrm{f}}{M_\mathrm{f}} \frac{M_\mathrm{in}}{L_\mathrm{pick} + L_\mathrm{in}} \Phi_\mathrm{pick} = \varepsilon \, \Phi_\mathrm{pick} \,, \tag{3.2}$$

where R_f is the feedback resistance, M_f is the feedback-coil mutual inductance, M_in is the mutual inductance between pickup and SQUID input coils, and L_pick, L_in are

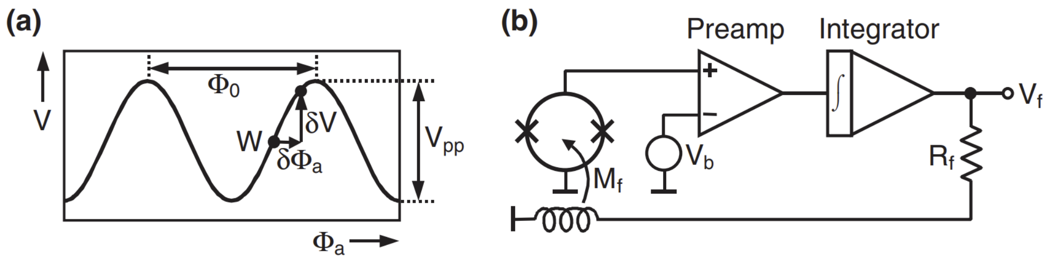

Figure 3.4: (a) V–Φ characteristic of dc SQUID showing the operating point W where the feedback voltage has approximately linear dependence on magnetic flux. (b) Direct-coupled flux-locked-loop (FLL) circuit with built-in pre-amplifier and integrator for maintaining the SQUID at the linear operating point.

the respective coil inductances. The conversion factor $\varepsilon \approx 0.549\,\mathrm{mV}/\Phi_0$, where Φ_0 is the magnetic flux quantum.

Table 3.1: Properties of the dc-SQUID sensor used in the measurements.

Pickup coil inductance (L_{pick})	553 nH
SQUID input coil inductance (L_{in})	400 nH
Mutual inductance (M_{in})	3.98 nH
Feedback-coil inductance (M_{f})	$22.83\,\Phi_0/\mathrm{mA}$
Feedback resistance (R_{f})	3 kΩ

The detection chain is characterized by the transfer coefficient β, defined as the ratio of the SQUID feedback-voltage amplitude $V_\perp$ to the transverse sample magnetization $\mu_0 M_\perp$:

$$\beta = \frac{V_\perp}{\mu_0\, M_\perp}\,. \tag{3.3}$$

The transfer coefficient was determined at each scan step from the feedback voltage following a $\pi/2$ pulse, for which $M_\perp = M_0$. The average value is $\beta \approx 4 \times 10^6\,\mathrm{V/T}$.

During ALP-search measurements the SQUID and FLL electronics are battery-powered to avoid electromagnetic interference from the mains. All other electronics (lock-in amplifier, field-ramp supply) are outside the Faraday cage; only shielded BNC cables penetrating the cage are used to carry signals.

The FLL output is fed into a Zurich Instruments MFLI Lock-in Amplifier (LIA), which demodulates the signal into in-phase (I) and quadrature (Q) channels. The complex output $I + iQ$ is recorded at a sampling frequency of 13.4 kHz and is used to generate power spectral densities (PSDs) via fast Fourier transformation (FFT).

3.5 Experimental control and data acquisition

Comprehensive centralized control of the CASPEr-Gradient-LF apparatus is implemented in Python to coordinate the complex timing requirements of pulsed-NMR experiments and manage communication between diverse hardware components. This control system enables synchronized operation of shim coils, rf pulse generation, and SQUID data acquisition.

The main functions of the control system include:

1. **Magnet and shim control:**
   a) Remote switching of shim power supply on and off
   b) Connection and disconnection of communication between power supply and shim coils
   c) Real-time setting of shim currents to desired values, with optional heater control for current stabilization
2. **Pulse generation and timing:**
   a) Command interface for injecting radiofrequency pulses
   b) Recording rf pulse timing via oscilloscope into HDF5 format
   c) Analysis of function generator data and pulse characterization
3. **Data acquisition and processing:**
   a) Recording SQUID data and automated creation of HDF5 files for storage
   b) Real-time analysis and monitoring of acquired signals
   c) Generation and analysis of power spectral densities

The control system orchestrates multiple measurement schemes including pulsed-NMR diagnostics for sample characterization, automated shimming procedures with feedback optimization, axion wind measurements, and magnetic field monitoring. Additionally, the system includes capabilities for NMR kinetic simulations, stochastic axion wind simulations, apparatus properties computation, and synchronized file management across multiple terminals.

3.6 Parameters summary

Table 3.2 summarizes the key parameters of the CASPEr-Gradient-LF apparatus for the methanol measurement reported in this thesis and for two projected future configurations using hyperpolarized samples. The thermal polarization of methanol is the dominant limiting factor in the current measurement. Hyperpolarized methanol or liquid ^{129}Xe could boost the polarization by several orders of magnitude, substantially improving the sensitivity.

3.7 Single-frequency sensitivity

The transverse magnetization M_1 induced by the ALP field couples to the pickup coil and SQUID via the pickup transfer coefficient α:

$$\alpha = \frac{\Phi}{\mu_0 M_1}, \tag{3.4}$$

where Φ is the magnetic flux in the SQUID and μ_0 is the vacuum permeability. Using Equations (2.9) and (2.11), the average ALP-induced flux power is

$$\langle \Phi_a^2 \rangle = \frac{1}{2}(\alpha\, u_n\, \mu_0 M_0 \Omega_a)^2 \begin{cases} \tau_a T_2^*, & \tau_a \ll T_2^* \\ \dfrac{\tau_a^2}{\pi}, & T_2^* \ll \tau_a \ll T_2 \\ T_2^2, & T_2 \ll \tau_a \end{cases}. \tag{3.5}$$

The signal amplitude in the power spectral density (PSD) is

$$S = \frac{\langle \Phi_a^2 \rangle}{\Gamma/2\pi}, \tag{3.6}$$

where Γ is the expected signal linewidth. Depending on which linewidth is narrower,

$$\Gamma = \begin{cases} \Gamma_a, & \Gamma_a \ll \Gamma_n \\ \Gamma_n, & \Gamma_a \gg \Gamma_n \end{cases}. \tag{3.7}$$

Within a spectral bin of width Γ, the number of PSD values available for averaging from a measurement of dwell time T is

$$N_\Gamma = \frac{\Gamma T}{2\pi} \gg 1\,, \tag{3.8}$$

and the noise standard deviation after averaging is

$$\sigma = \sqrt{\frac{2\pi}{\Gamma T}}\,\sigma_0\,, \tag{3.9}$$

where σ_0 is the single-bin noise level of the SQUID system.

Using a detection threshold of $S \geq 3.355\,\sigma$—corresponding to a true signal $5\,\sigma$ above the noise detected at 95% confidence [4]—the ALP coupling strengths excluded when no signal is found satisfy

$$|g_{\mathrm{aNN}}| \geqslant \begin{cases} \eta\, T^{-1/4}\,(T_2^*)^{-3/4}\,\tau_a^{-1/2}, & \tau_a \ll T_2^* \\ \eta\,\pi^{-1/4}\,T^{-1/4}\,(T_2^*)^{-1}\,\tau_a^{-1/4}, & T_2^* \ll \tau_a \ll T_2 \\ \eta\,\pi^{-3/4}\,T^{-1/4}\,T_2^{-1}\,(T_2^*)^{-1}\,\tau_a^{3/4}, & T_2^* \ll T_2 \ll \tau_a \\ \eta\,\pi^{-3/4}\,T^{-1/4}\,T_2^{-2}\,\tau_a^{3/4}, & T_2^* = T_2 \ll \tau_a \end{cases}\,, \tag{3.10}$$

where the prefactor η collects the apparatus-dependent constants:

$$\eta = \frac{2\sqrt{3.355\,\sigma_0}}{\pi^{1/4}\,\alpha\,\mu_0\,M_0\,v_\perp\,\sqrt{\hbar c\,\rho_{DM}}}\,. \tag{3.11}$$

Larger magnetization M_0 and lower noise σ_0 both improve η and hence the sensitivity.

3.8 Scanning procedure

The CASPEr-Gradient-Low-Field experiment scans the ALP-mass parameter space by sweeping the magnitude of $\mathbf{B}_0$, which tunes the Larmor frequency of the sample to a series of values $\nu_i \in [\nu_{\mathrm{start}}, \nu_{\mathrm{end}}]$. One step of the scan consists of:

A. Pulsed-NMR measurement to determine the resonance frequency and the effective linewidth (T_2^*).

B. Spin-echo measurement to determine T_2.

C. Free-precession data acquisition with the SQUID magnetometer—no transverse pulse applied—to record the ALP-signal channel over the dwell time T.

D. Fourier transform of the magnetometer data to produce a power spectrum.

E. Binning of PSD values within frequency bins of width Γ, and averaging to reduce noise.

F. Statistical analysis of binned PSD values (e.g., histogram comparison with the expected noise distribution) to identify DM-candidate frequencies.

G. If a candidate is found: repeat the measurement at the same field value and perform additional checks.

H. If no candidate is found: record the exclusion contour for this frequency step and ramp $\mathbf{B}_0$ to the next value.

The procedure is illustrated in Figure 3.5. A significant excess surviving repeated measurement can be tested further using the daily and annual modulation signatures of ALP dark matter [18].

The frequency-step size Γ_{scan} is set by the sensitive bandwidth of a single step. An experiment is sensitive to signals within a bandwidth equal to the maximum of Γ_n and Γ_a around a given Larmor frequency. Choosing a step size larger than Γ_{scan} leaves gaps; choosing a smaller step size multiplies the number of steps without improving coverage.

3.9 Scanning optimization

With fixed total measurement time T_{tot}, the dwell time can in principle depend on frequency: $T \propto \nu^n$. Maximizing the sensitive area in the $\log \nu$–$\log |g_{\mathrm{aNN}}|$ parameter space over the full scan determines the optimal exponent n. The derivation is given in Appendix C; the result is $n = 0$ (uniform dwell time) for regimes (1)–(3) and $n = -1$ (dwell time decreasing as $1/\nu$) for regime (4).

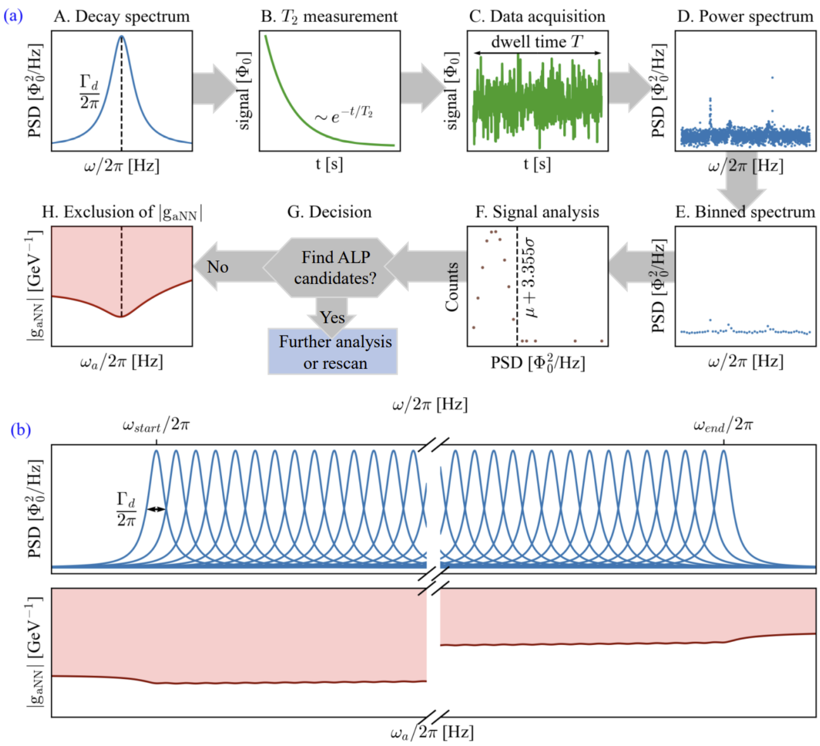

Figure 3.5: Schematic of the experimental procedure. (a) One step of the scan: pulsed-NMR diagnostics, free-precession data acquisition, PSD computation and binning, and statistical analysis using a histogram of PSD values. (b) Example decay spectra (blue curves) and the resulting sensitivity region in the ALP mass-coupling space (reddish area) after scanning over a frequency range. Reproduced from Ref. [32].

With optimal dwell-time allocation, the scanning sensitivity is

$$|g_{\mathrm{aNN}}| \geqslant \begin{cases} \eta\left(\dfrac{T_{\mathrm{tot}}}{Q_a \ln(\nu_{\mathrm{end}}/\nu_{\mathrm{start}})}\right)^{-1/4} \times (T_2^*)^{-3/4}\, \tau_a^{-1/2}, & \tau_a \ll T_2^* \text{ or } T_2^* \ll \tau_a \ll T_2 \\ \eta\left(\dfrac{\pi^2 T_{\mathrm{tot}}}{Q_a \ln(\nu_{\mathrm{end}}/\nu_{\mathrm{start}})}\right)^{-1/4} \times (T_2^*)^{-3/4}\, T_2^{-1}\, \tau_a^{1/2}, & T_2^* \ll T_2 \ll \tau_a \text{ or } T_2^* = T_2 \ll \tau_a \end{cases} . \tag{3.12}$$

Several conclusions follow directly:

- **Time scaling.** The sensitivity scales as $T_{\mathrm{tot}}^{-1/4}$ in all regimes, a consequence of the long-dwell-time limit $T \gg \tau_a, T_2^*, T_2$ [5, 9]. An order-of-magnitude increase in measurement time improves the exclusion bound by only a factor $10^{1/4} \approx 1.8$.

- **Effect of T_2^*.** Longer T_2^* is beneficial in all regimes. When $\tau_a \ll T_2^*$, it increases the r.m.s. tipping angle and narrows the signal linewidth. When $T_2^* \ll \tau_a$, it increases the fraction u_n of spins on resonance.
- **Effect of T_2.** Longer T_2 improves sensitivity in the regime $T_2 \ll \tau_a$, where T_2 directly limits the r.m.s. tipping angle.

The sensitivity plots computed with uniform ($n = 0$) and frequency-adaptive ($n = -1$) dwell-time allocation for an example scan range $\nu_{\text{end}}/\nu_{\text{start}} = 10$ are shown in Figure 3.6. The area covered in the log ν–log $|g_{\text{aNN}}|$ plane is larger with optimal n.

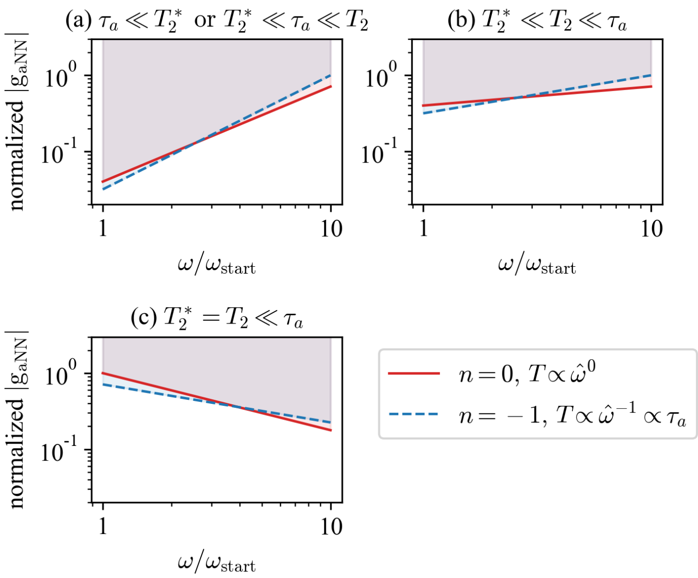

Figure 3.6: Sensitivity plots on a log-log scale for two dwell-time allocation schemes: uniform ($n = 0$) and frequency-adaptive ($n = -1$). The shaded regions indicate the parameter space accessible to the experiment. The coupling strength is normalized in each regime so that the highest point equals 1. The scan range is $\nu_{\text{end}}/\nu_{\text{start}} = 10$. Reproduced from Ref. [32].

In summary, the optimal strategy for a frequency-scanning ALP search is to work at the maximum achievable T_2 and T_2^*, use a step size equal to the sensitive bandwidth, and distribute the total measurement time uniformly across frequency steps (or as $T \propto \nu^{-1}$ in regime (4)).

Table 3.2: Parameters of the CASPEr-Gradient-LF apparatus for the current measurement (thermally polarized methanol) and two projected future configurations. Polarization is given as molar polarization (polarization×concentration).

	This work	Examples of future CASPEr-Gradient-LF	
Sample	Methanol	Methanol[a]	Liquid ^{129}Xe[b]
Spin density (cm^{-3})	5.9×10^{22}	5.9×10^{22}	$\sim 10^{22}$
Sample volume (cm^3)	1.2	1.2	~ 1
Polarization technique	Thermal	BF[c]/ PHIP[d]/ DNP[e]	SEOP[f]
Polarization	1.8×10^{-7}	$> 1.1 \times 10^{-5}$	~ 1
T_2 (s)	0.72	1	1000
B_0 inhomogeneity (ppm)	10	≤ 2	≤ 2
Electronic noise ($\mu\Phi_0/Hz^{1/2}$)	7.1	1.0	1.0
Flux of SPN[g] ($\mu\Phi_0$)	0.39	0.39	0.16
Scan range	1.348 450 MHz–1.348 690 MHz	~1 kHz–4.3 MHz	~1 kHz–1.2 MHz
$\lvert g_{aNN} \rvert$ sensitivity (GeV^{-1})	3×10^{-2}	1.0×10^{-6}–1.2×10^{-5}	10^{-12}–10^{-11}

[a] Methanol at 180 K prepolarized at 2 T; measurement time 100 τ_a per scan step.
[b] ^{129}Xe-enriched liquid sample.
[c] Brute-force prepolarization in a strong field at low temperature.
[d] Parahydrogen-induced polarization (PHIP).
[e] Dynamic nuclear polarization (DNP).
[f] Spin-exchange optical pumping (SEOP).
[g] Spin-projection noise (SPN).

4 Results

Sections 4.1–4.3 are adapted from Ref. [28].

The scanning strategy developed in Chapter 3 is applied here to the proof-of-concept methanol search. This chapter presents the measurement, data-analysis pipeline, candidate evaluation, and resulting exclusion limits.

4.1 The methanol measurement

The scanning framework developed in Sections 3.7–3.9 was applied in a proof-of-concept ALP search using the CASPEr-Gradient-LF apparatus with a thermally polarized methanol sample. This section describes the measurement procedure and parameters. In the following, g_{ap} denotes the ALP-proton gradient coupling constant, i.e. g_{aNN} evaluated for proton spins.

4.1.1 Measurement parameters

The search covered a 240 Hz band centered at 1.348 570 MHz, corresponding to proton Larmor frequencies driven by a bias field $|\mathbf{B}_0| \approx 31.67\,\mathrm{mT}$. This maps to ALP masses in the range 5.576741–$5.577733\,\mathrm{neV}/c^2$. The frequency step size was 6 Hz, approximately half the NMR linewidth, so that the resonant windows of adjacent steps overlap. Three sequential scans over the same range were performed. The most sensitive scan was treated as the primary scan and the other two (scans A and B) as consistency checks.

The ALP coherence time at the search frequency is $\tau_a \approx 0.35\,\mathrm{s}$, obtained from Equation (2.16). The measured relaxation times are $T_2^* = 24.2(4.0)\,\mathrm{ms}$ and $T_2 = 0.72(5)\,\mathrm{s}$. Since $T_2^* \ll \tau_a \lesssim T_2$, the measurement falls in regime (2) of Table 2.1. The spectral factor is approximately $u \approx T_2^*/\tau_a \approx 7\,\%$.

4.1.2 Step sequence

At each frequency step the following actions were performed in order:

1. Pulsed-NMR measurement (a $\pi/2$ pulse followed by free-induction decay) to determine the Larmor frequency, linewidth $\Delta\nu$ (and hence $T_2^* = 1/(\pi\Delta\nu)$), and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters and calibrates the sensitivity at each step.

2. Carr-Purcell-Meiboom-Gill (CPMG) [10] pulse sequence prior to the first scan step to determine T_2 from the exponential decay of spin-echo amplitudes.

3. 5 s dwell time to allow ring-down noise to decay and the magnetization to return close to its equilibrium value along $\mathbf{B}_0$.

4. 100 s data acquisition without any pulses, recording the SQUID feedback voltage V_f at 13.4 kHz sampling rate. This window corresponds to approximately 280 ALP coherence times and is the ALP-search channel.

5. Frequency step: the Larmor frequency is increased by ≈6 Hz by ramping the Helmholtz coils or the superconducting magnet.

6. Another 5 s dwell time after the field ramp before repeating.

The PSDs were recorded after demodulation of V_f by the lock-in amplifier; its reference frequency was detuned by 820 Hz from the Larmor frequency so that the NMR signal appears at a non-zero difference frequency, making it easily distinguishable from noise. Example PSD spectra and NMR characterization plots are shown in Figure 4.1.

Due to Earth's rotation, the component of the ALP-field gradient perpendicular to $\mathbf{B}_0$ varies over the course of the measurement. The resulting daily modulation of the signal amplitude is predictable and was tracked over the approximately six-hour duration of the scans.

4.2 Data analysis

Each 100 s time series is converted to a PSD and processed through three successive steps.

4.2.1 Spike suppression

A narrow Savitzky-Golay (SG) filter with a window half-width equal to half the expected ALP linewidth at the scanned frequency is applied to the PSD. The residual between the raw PSD and the filter output is fitted with a χ^2 distribution with two degrees of freedom. Bins that deviate from the filter response by more than the fitted threshold, together with their four nearest neighbors, are replaced by the local filter average. This step suppresses narrow EMI spikes while preserving potential ALP signals. In the primary scan, only 0.002 % of bins were modified in this way.

4.2.2 Baseline removal

A reduced chi-square test on the spike-corrected spectrum reveals a non-flat baseline. A second, broad SG filter with a window of 200 ALP linewidths is applied to estimate this baseline. The spike-corrected spectrum is then normalized by subtracting and dividing by this baseline, producing a spectrum with a flat expected noise floor.

4.2.3 Matched filtering and NPE

A matched filter using the expected ALP signal lineshape $\lambda_\perp(\nu, \nu_a)$—the normalized spectral lineshape of the ALP-field gradient component perpendicular to $\mathbf{B}_0$, as given in Ref. [18]—is convolved with the normalized PSD. The result, normalized by its standard deviation, is the normalized power excess (NPE). The matched-filter step size is half the ALP linewidth.

4.2.4 Hypothesis test and rescan threshold

Frequency bins where the NPE exceeds the rescan threshold Θ_{RE} are flagged as outliers. The threshold was determined via Monte Carlo simulations of the NPE distribution under the null hypothesis that a $5\,\sigma$ ALP signal is present, yielding $\Theta_{\mathrm{RE}} = 3.43$ at the 95 % confidence level (corresponding to a p-value of 0.001).

The overall analysis efficiency, accounting for the effect of baseline removal on signal recovery, was determined from simulations to be $\eta_{\mathrm{rec}} = 88.2(4.6)\,\%$. This efficiency is applied when converting NPE values to coupling-constant limits.

4.3 Exclusion results

4.3.1 Candidates and consistency checks

Table 4.1 lists the outlier counts and ALP signal candidates from all three scans. The total number of outliers in each scan exceeds the expectation for white noise, owing to non-white-noise structures that are neither narrow nor broad enough to be removed by the SG filters. By comparing the resonant window (within ±5 NMR linewidths of the Larmor frequency) with off-resonance background, and by cross-checking the primary scan against scans A and B, these noise-induced outliers can be identified.

Table 4.1: Results of the three ALP scans. N_{full}: total NPE bins. X_{full}: outliers above rescan threshold. $\langle X_{\mathrm{full}} \rangle$: expected value assuming white noise. N_{res}: bins in resonant windows. X_{res}: ALP signal candidates. $\langle X_{\mathrm{res}} \rangle$: expected value. X'_{res}: candidates after merging those within one ALP linewidth. X''_{res}: candidates in this scan also appearing in the primary scan within four ALP linewidths. Parenthetical fractions give the fraction of N_{full}.

	Primary scan	Scan A	Scan B
N_{full}	481 680	481 680	481 680
X_{full}	6393 (1.32 %)	5760 (1.20 %)	6202 (1.29 %)
$\langle X_{\mathrm{full}} \rangle$	145 (0.03 %)	145 (0.03 %)	145 (0.03 %)
N_{res}	8430	8430	8430
X_{res}	5	7	3
$\langle X_{\mathrm{res}} \rangle$	3	3	3
X'_{res}	5	5	3
X''_{res}	—	0	0
$\langle X''_{\mathrm{res}} \rangle$	—	0	0

No candidate from the primary scan was found in either of the consistency-check scans within four ALP linewidths. The search therefore yields no statistically significant evidence for ALP dark matter at the detection sensitivity.

4.3.2 Coupling-constant limit

$$|g_{\mathrm{aNN}}| \leq \sqrt{\frac{e_5}{\eta_{\mathrm{rec}} P_{\perp,1}}}, \tag{4.1}$$

In the absence of a detection, an upper limit on the ALP-proton gradient coupling is obtained from

$$|g_{\mathrm{ap,\,lim}}| = \sqrt{\frac{e_5}{\eta_{\mathrm{rec}}\, P_{\perp,1}}}\,, \tag{4.2}$$

where e_5 is the signal power corresponding to the $5\,\sigma$ NPE threshold, $P_{\perp,1}$ is the reference signal power for $|g_{\mathrm{ap}}| = 1\,\mathrm{GeV}^{-1}$, and η_{rec} is the analysis efficiency defined above. The resulting exclusion limits are plotted in Figure 4.2.

The measurement sets the limit $|g_{\mathrm{ap}}| \lesssim 3 \times 10^{-2}\,\mathrm{GeV}^{-1}$ over the scanned ALP mass range 5.576741–$5.577733\,\mathrm{neV}/c^2$. This is the first dedicated laboratory search in this mass range. The sensitivity is dominated by the small thermal polarization of the methanol sample; hyperpolarized samples are expected to improve it by several orders of magnitude.

4.4 Earth-bound ALP measurement

4.4.1 Procedure and parameters

A data taking with thermally polarized methanol took place at Dec 14, 2022. The measurement scanned

$$1.348\,585\,\mathrm{MHz} \pm 24\,\mathrm{Hz}$$

by sweeping the magnitude of the bias field $|\mathbf{B}_0| \approx 31.674\,03\,\mathrm{mT}$. This corresponds to an ALP mass range of

$$5.577\,299\,\mathrm{neV/c^2} \pm 50\,\mathrm{feV/c^2}\,.$$

The scanning took 8 steps. In each scan step, the following actions were performed in order:

1. Set sweep coil current.

2. 1-Pulse measurement (a $\pi/2$ pulse followed by free decay signal) to determine the Larmor frequency, linewidth $\Delta\nu$ (and hence $T_2^* = 1/(\pi\Delta\nu)$), and NMR signal power. A Lorentzian fit to the resulting PSD peak extracts these parameters and calibrates the sensitivity at each step.

3. Carr-Purcell-Meiboom-Gill (CPMG) [10] pulse sequence prior to the first scan step to determine T_2 from the exponential decay of spin-echo amplitudes.

4. A 1-hour ALP measurement without any pulses.

5. 1-Pulse measurement like step 2.

6. CPMG measurement like step 3.

Delays ($\gg T_1$) were inserted between pulsed-NMR measurements to allow the magnetization to return close to its equilibrium value along $\mathbf{B}_0$. A schematic of the measurement procedure is shown in Figure 4.3. Note that the 1-hour ALP measurement duration is much longer than the 100-second duration used in the data taking described in Section 4.1. The long measurement duration results in a much narrower PSD bin width, which is beneficial for the sensitivity to ALP signals with long coherence time, such as the Earth-bound ALP halo. A summary of the measurement parameters is given in Table 4.2. Long measurement durations make this measurement sensitive to the long-coherence-time ALP fields. Therefore, this measurement is chosen to demonstrate the sensitivity of the CASPEr-Gradient-LF apparatus to the Earth-bound ALP halo model.

Table 4.2: Technical parameters of the Earth-bound ALP measurement (December 14, 2022). Here SNR indicates the signal-to-noise ratio of the free decay signal after a $\pi/2$ pulse.

Parameter	Name or value
Sample	thermally polarized methanol
Number of scan steps	8
Sweep coil current increment	0.8 mA
Frequency increment per step	6 Hz
Frequency scan range	1.348 585 MHz ± 24 Hz
NMR linewidth (FWHM) in one step	≈ 10 Hz
T_2 relaxation time	≈ 1 s
Signal-to-noise ratio (SNR)	≈ 120
Axion measurement duration per step	1 h

4.4.2 Expected signal and sensitivity

Due to the daily rotation of the Earth, the component of the ALP-field gradient perpendicular to $\mathbf{B}_0$ varies over the course of the measurement. Using the ALP signal model established in Chapter 2.3, the expected signal amplitude can be calculated. In Figure 4.4, the evolution of the expected Rabi frequency of the pseudomagnetic field is plotted (assuming all ALPs are at $2p$ state). Since the Rabi frequency $\Omega_a \approx 2\pi \times 1\,\mathrm{mHz}$, the tipping angle of the magnetization can be estimated as $\theta \approx \Omega_a T_2 \approx 0.002\,\pi$.

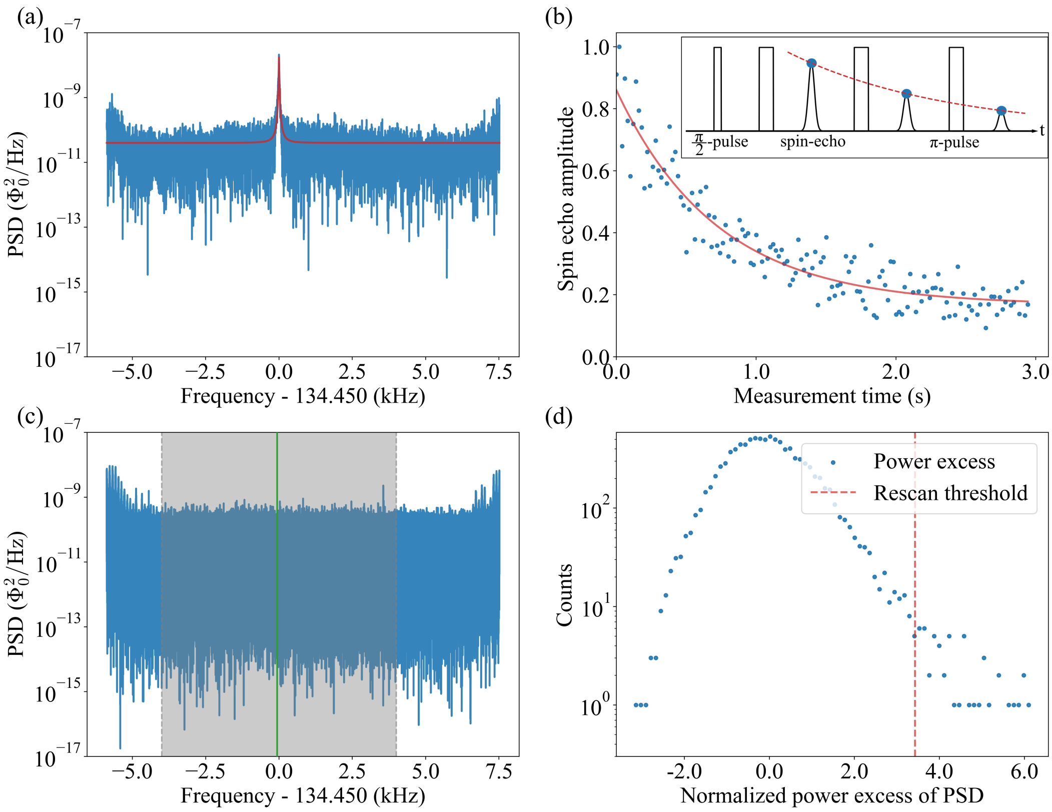

Figure 4.1: (a) PSD after a $\pi/2$ pulse. The NMR peak is fitted with a Lorentzian (red line); the center is at 1 348 449.21(5) Hz with a FWHM of 14.79(1) Hz. **(b)** CPMG spin-echo amplitudes (blue dots) and exponential fit (red), giving $T_2 = 0.72(5)$ s. **(c)** PSD of a 100 s ALP-search step. The gray analysis window (8 kHz wide) excludes filter artifacts near the band edges; the green resonant window (±5 NMR linewidths = ±70 Hz) is where ALP signal candidates are sought. **(d)** Histogram of normalized power excess (NPE) derived from (c), with the rescan threshold Θ_{RE} (red line) at the 95 % confidence level for a 5 σ ALP detection. Reproduced from Ref. [28].

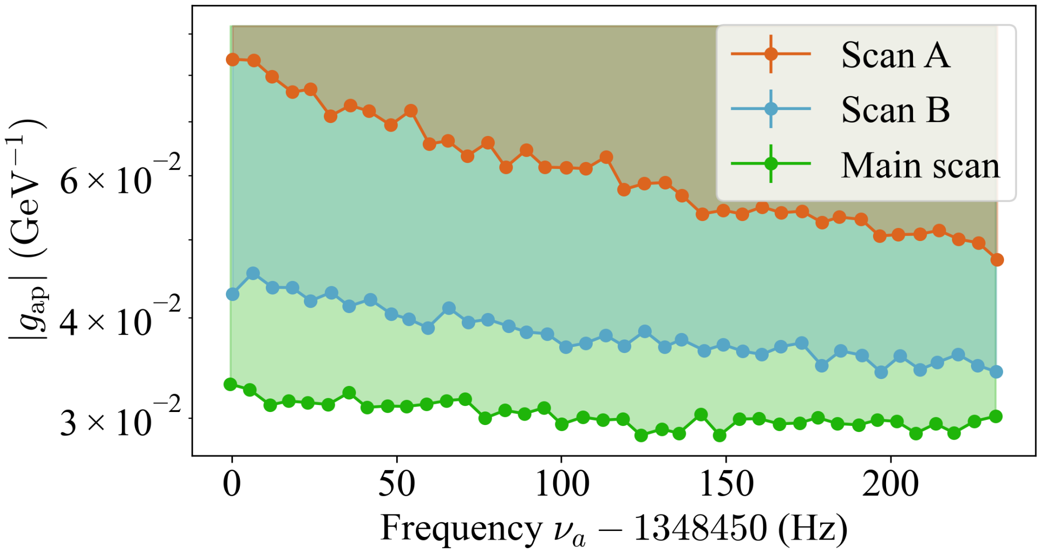

Figure 4.2: ALP gradient-proton coupling excluded at 95 % confidence level by the three sequential scans. Each data point represents the sensitivity for one scan step. The variation in sensitivity between scans reflects the daily modulation of the ALP wind direction relative to the bias field. Reproduced from Ref. [28].

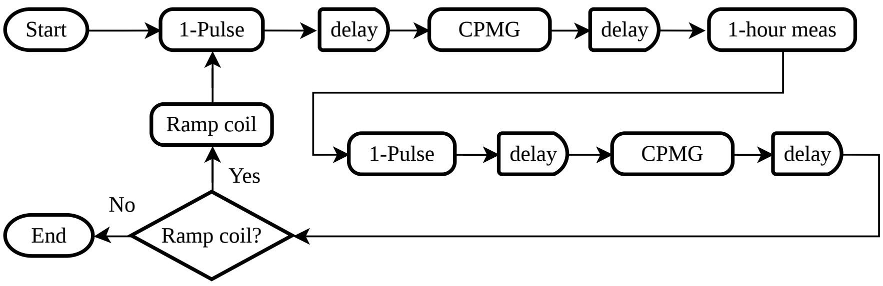

Figure 4.3: Schematic of the Earth-bound ALP measurement procedure.

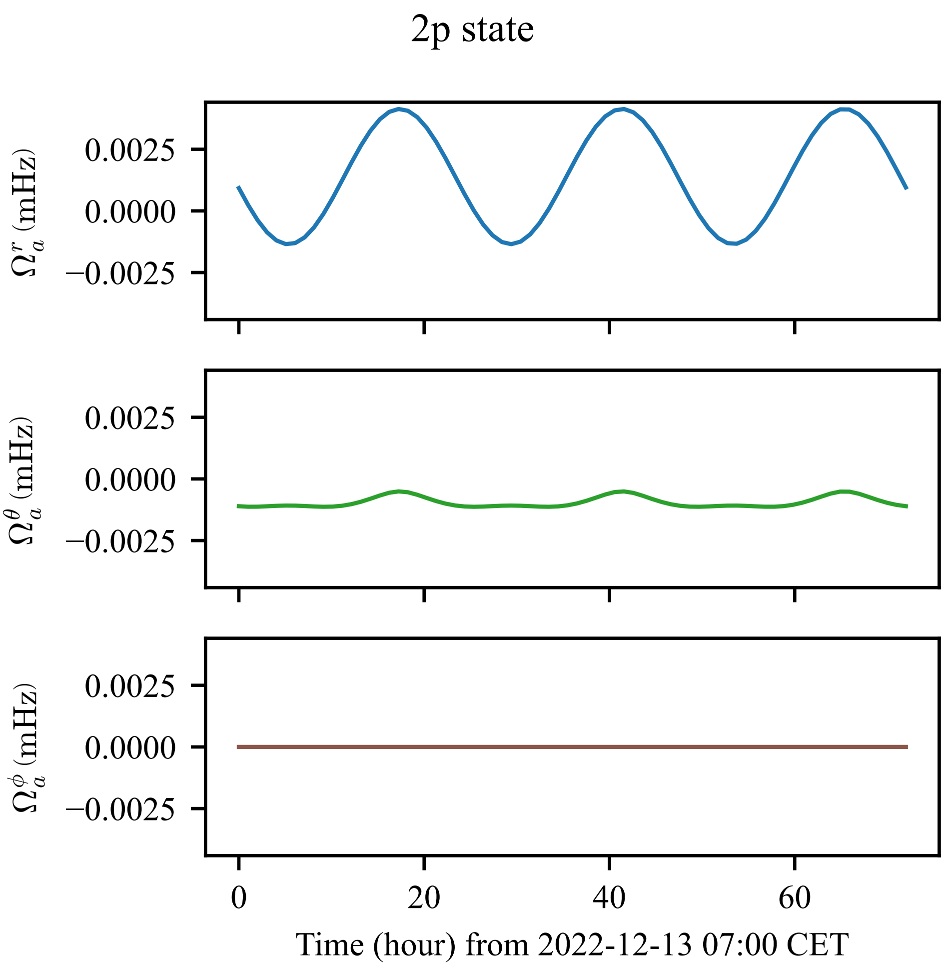

Figure 4.4: Expected evolution of the ALP-induced Rabi frequency (pseudomagnetic field) over one sidereal day for the Earth-bound ALP halo model. The amplitude modulation is driven by the daily rotation of the laboratory frame relative to the ALP-field gradient; the plotted curve shows the combined effect of the geometry of $\mathbf{B}_0$ and the predicted halo profile.

5 Conclusions & Outlook

5.1 Summary of results

This thesis investigated the use of nuclear magnetic resonance for searches for axionlike dark-matter particles through the nuclear-gradient coupling, focusing on the CASPEr-Gradient-Low-Field experiment at Mainz.

The theoretical analysis in Chapter 2 and in Sections 3.7–3.9 of Chapter 3 (adapted from Ref. [32]) established the following key results:

- **Optimal step size.** The frequency-step size for scanning should equal the sensitive bandwidth of a single measurement, namely the larger of the NMR linewidth Γ_n and the ALP linewidth Γ_a.
- **Optimal relaxation times.** The transverse relaxation times T_2 and T_2^* should be maximized. Longer T_2^* is beneficial in all regimes: it increases either the r.m.s. tipping angle or the fraction of on-resonance spins. Longer T_2 improves sensitivity when $T_2 \ll \tau_a$.
- **Sensitivity scaling.** The sensitivity to $|g_{\mathrm{aNN}}|$ scales as $T_{\mathrm{tot}}^{-1/4}$ with total measurement time in all regimes considered ($T \gg \tau_a, T_2^*, T_2$). This slow scaling means improving T_2, T_2^*, or the detector noise σ_0 is more efficient than simply accumulating more measurement time.
- **Optimal dwell-time allocation.** For regimes (1)–(3), the dwell time should be distributed uniformly across frequency steps ($T \propto \nu^0$). For regime (4) ($T_2^* = T_2 \ll \tau_a$), the optimal strategy is $T \propto \nu^{-1}$.
- **Four regimes.** The parameter space is divided into four regimes by the relative magnitudes of T_2, T_2^*, and τ_a. Each regime has a distinct signal amplitude and optimal dwell time, as summarized in Table 2.1.

These results were applied in a proof-of-principle ALP search with the CASPEr-Gradient-LF apparatus (adapted from Ref. [28]), described in Sections 4.1–4.3 of Chapter 4:

- **First laboratory search in the target mass range.** A thermally polarized methanol sample was used to search for the ALP-proton gradient coupling in the mass range 5.576741–5.577733 neV/c^2 (Compton frequency 240 Hz around 1.348 570 MHz). This is the first dedicated laboratory search in this mass range.

- **No dark-matter candidates found.** Three sequential scans were performed. No ALP signal candidates survived the cross-scan consistency check, yielding the constraint $|g_{\mathrm{ap}}| \lesssim 3 \times 10^{-2}\,\mathrm{GeV}^{-1}$ at 95 % confidence level.

- **Methodology validated.** The data-analysis pipeline—Savitzky-Golay spike suppression, baseline removal, matched filtering, and NPE hypothesis test—was validated through synthetic-signal injection. The overall analysis efficiency is $\eta_{\mathrm{rec}} = 88.2(4.6)\,\%$.

- **Sensitivity limited by thermal polarization.** The proton thermal polarization at 180 K and 31.67 mT is only $\approx 1.8 \times 10^{-7}$, which dominates the sensitivity budget. The apparatus is otherwise ready for more sensitive searches with hyperpolarized samples.

5.2 Outlook

The framework developed here can be extended in several directions:

- **Hyperpolarized samples.** Hyperpolarized ^{129}Xe or methanol can increase the polarization by six or more orders of magnitude relative to thermal polarization, directly improving M_0 and hence the sensitivity prefactor η in Equation (3.11). The T_1 relaxation of hyperpolarized samples limits the measurement window and modifies the optimal dwell-time strategy. Examples of projected sensitivities are given in Table 3.2.

- **Seeded measurements.** Applying a transverse oscillating field at the resonance frequency ("seeding") to heterodyne the ALP signal was outside the scope of this work. Seeding could improve sensitivity by coherently amplifying the ALP-induced tipping angle [5].

- **Improved field homogeneity and longer T_2.** Reaching the design inhomogeneity of 2 ppm over the sample volume would lengthen T_2^* and increase the fraction of on-resonance spins. Combined with samples having long intrinsic T_2, this improvement is beneficial in all four regimes.

- **High-field extension.** The CASPEr-Gradient-High-Field experiment will operate with a 14.1 T magnet, reaching Larmor frequencies up to 600 MHz and accessing higher ALP masses. The scanning framework derived here applies directly to that regime.

- **Low-frequency regime.** Below about 1 kHz, τ_a can exceed 1000 s, and the assumption $T \gg \tau_a$ breaks down. An adaptive scanning strategy that accounts for the long coherence time will be required for CASPEr-ZULF [16] and related low-frequency experiments.

Data and source code availability

The LaTeX source of this dissertation, including all analysis scripts and figures, is publicly available in the GitHub repository of the CASPEr Collaboration:

https://github.com/CASPEr-Collaboration/ZhangYuzhe-Dissertation

The numerical simulation package `axionbloch` used in Appendix A is available as open-source software [30]:

https://github.com/Yuzhe98/AxionBloch

The raw experimental data underlying the methanol ALP search presented in Chapter 4 are associated with Ref. [28]. Requests for access to the raw data should be directed to the corresponding authors of that publication.

The scanning-optimization analysis underlying Sections 3.7–3.9 is associated with Ref. [32].

A Numerical simulation with axionbloch

This appendix is adapted from Ref. [31].

Analytical estimates of the ALP-induced spin-precession signal (Chapter 2) are valid in the long-dwell-time limit $T \gg \tau_a, T_2^*, T_2$. At lower Compton frequencies, the ALP coherence time becomes comparable to or longer than experimentally accessible dwell times, and the scaling laws no longer apply. Numerical simulation is then the tool of choice. This appendix describes `axionbloch`, an open-source Python package that simulates spin dynamics under both magnetic and axion-induced pseudomagnetic fields.

A.1 Software architecture

The package uses an object-oriented design separating three concerns: experimental configuration (`Sample`, `Magnet`), field generation (`MagField`), and simulation management (`Simulation`, `Simulations`). Axion-field models are encapsulated in dedicated classes (e.g., `MilkyWayAxionHalo`), allowing new models to be added without modifying the simulation framework. The architecture is illustrated in Figure A.1.

A.2 Calibration

Three standard NMR experiments are used to benchmark the simulation against known analytical results.

A.2.1 Pulsed NMR: free decay and spin echo

After a 90° pulse, the magnetization tips into the transverse plane and decays with the dephasing time T_2^* (free-induction decay). A subsequent 180° pulse refocuses the inhomogeneous dephasing, producing a spin echo whose amplitude decays with the true T_2. The simulated signals reproduce both behaviors, calibrating the pulse-induced rotation, Larmor precession, and transverse relaxations. The simulated magnetization is shown in Figure A.2 (a).

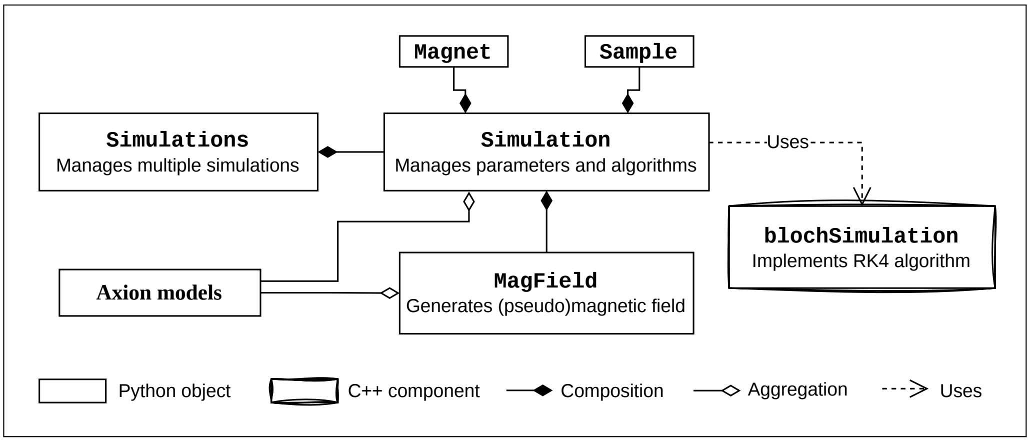

Figure A.1: Architecture of `axionbloch`. Each `Simulation` instance receives parameters from `Magnet`, `Sample`, and `MagField`. For axion simulations, an axion-model object is required both to configure the simulation and to generate the pseudomagnetic field. Numerical integration is performed via the `blochSimulation` API, which provides Python access to the C++ backend. Reproduced from Ref. [31].

A.2.2 Continuous-wave NMR

For a weak oscillating field on resonance ($\gamma B T_2^* \ll 1$), the transverse magnetization grows as $\gamma B T_2^*(1 - e^{-t/T_2^*})$ [2]. The simulation reproduces this analytical form, as shown in Figure A.2 (b). This benchmark calibrates both the Larmor precession and the T_2^* relaxation under continuous driving—the same physical regime as the ALP-induced signal.

A.2.3 Hyperpolarized sample and T_1 relaxation

Hyperpolarization boosts the initial spin polarization above its thermal value. In the simulation, the initial polarization is set as a parameter of the `Sample` class. A sample polarized significantly above thermal equilibrium relaxes toward M_0 with the longitudinal time constant T_1. The simulated $M_z(t) = M_0[1 - (M_0^{\text{init}}/M_0 - 1)e^{-t/T_1}]$ reproduces the expected exponential approach to equilibrium, as illustrated in Figure A.3.

A.3 Axion simulation example

As a demonstration, the `MilkyWayAxionHalo` model simulates the pseudomagnetic field from virialized galactic dark matter following the standard halo model

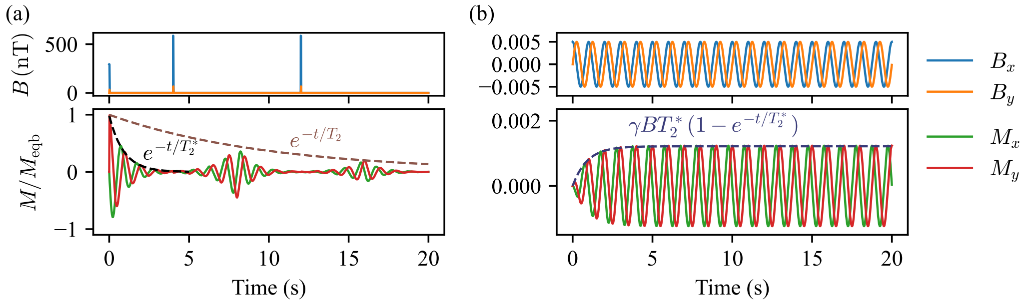

Figure A.2: Calibration of `axionbloch` against analytical NMR results. Upper rows: applied transverse magnetic fields; lower rows: normalized simulated magnetization. **(a)** After 90° and 180° pulses, the simulation produces free-induction decay (timescale T_2^*) and a spin echo (timescale T_2). **(b)** Under a weak continuous-wave drive on resonance, the transverse magnetization grows as $\gamma B T_2^*(1 - e^{-t/T_2^*})$ (dashed line). Reproduced from Ref. [31].

[15, 18]. The axion field oscillates at the Compton frequency with a Maxwell-Boltzmann velocity distribution setting the spectral linewidth. The stochasticity of the field amplitude is accounted for by generating many independent realizations of the pseudomagnetic field.

Figure A.4 shows results for a ^{129}Xe sample polarized to 50 %. A single realisation (left) illustrates the stochastic fluctuations in both the pseudomagnetic field and the induced transverse magnetization. An ensemble of 1000 realizations (right) yields the mean signal and its standard deviation band. The signal initially grows as the ALP field drives the magnetization away from equilibrium, and then decays on the T_1 timescale as the hyperpolarization is lost.

These simulations validate the three-regime picture in Equation (2.11): at low frequencies where $\tau_a \gg T$, the signal grows linearly in dwell time; in the intermediate regime $\tau_a \sim T$, the signal departs from the analytical long-dwell-time scaling; and at high frequencies where $T \gg \tau_a$, the analytical results apply directly.

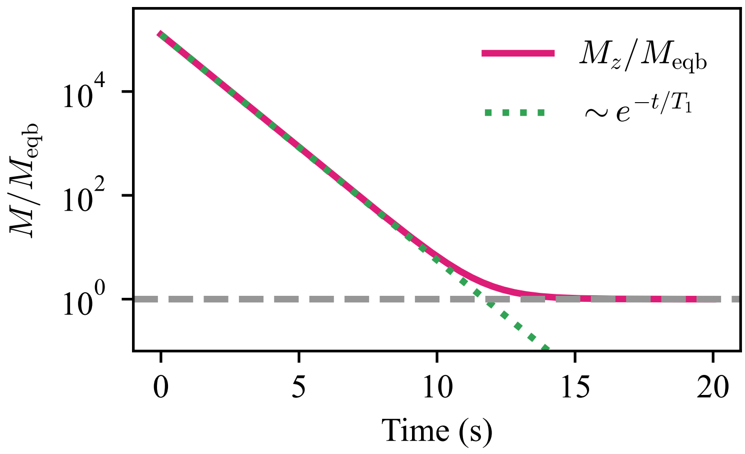

Figure A.3: Simulated T_1 relaxation of a hyperpolarized sample in a static bias field. The axial magnetization M_z is normalized by the equilibrium magnetization M_0. The exponential approach to equilibrium calibrates the longitudinal relaxation in the simulation. Reproduced from Ref. [31].

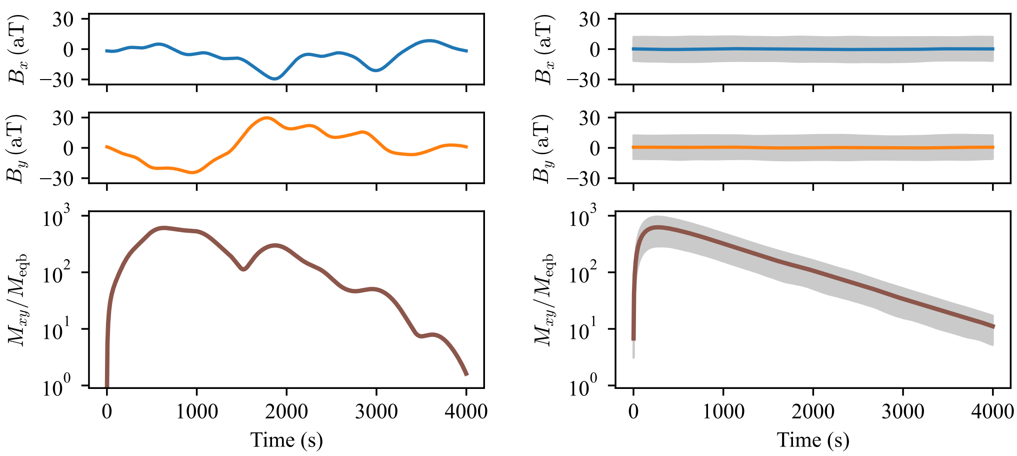

Figure A.4: Simulated axion-induced NMR signals for a virialized Milky Way axion dark matter halo. Upper panels: pseudomagnetic fields B_x and B_y in the transverse plane. Lower panels: amplitude of the induced transverse magnetization M_{xy}, normalized by M_0. Left: single simulation realisation. Right: ensemble of 1000 realizations; solid lines show the mean and shaded bands show $\pm\sigma$. The sample is ^{129}Xe initially polarized to 50 %. Reproduced from Ref. [31].

B Statistical hypothesis test for ALP detection

B.1 The Neyman–Pearson Lemma

The *Neyman–Pearson lemma* is a foundational result in statistical hypothesis testing. It establishes the form of the most powerful test for discriminating between two simple hypotheses at a fixed significance level.

B.1.1 Problem setup

Suppose we observe data x drawn from a probability distribution. We wish to test

$$H_0 : x \sim f_0(x)$$

against

$$H_1 : x \sim f_1(x),$$

where both $f_0(x)$ and $f_1(x)$ are fully specified probability density functions. The goal is to construct a test that maintains a fixed false-alarm probability (Type I error) while maximizing the probability of correctly detecting the alternative hypothesis (power).

B.1.2 Likelihood ratio and the lemma

The key quantity is the likelihood ratio

$$\Lambda(x) = \frac{f_1(x)}{f_0(x)}.$$

For a fixed significance level α, the Neyman–Pearson lemma states that the most powerful test rejects H_0 when $\Lambda(x) > k$, where k satisfies $P(\Lambda(x) > k \mid H_0) = \alpha$. The likelihood-ratio test is therefore optimal among all tests with the same false-positive rate.

B.1.3 Type I error and power

The significance level $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$ is the false-alarm probability. The power $1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ true})$ is the probability of correctly detecting an ALP signal. The Neyman–Pearson lemma guarantees that the likelihood-ratio test maximizes the power for fixed α.

B.1.4 Application to the ALP search

In the ALP data analysis described in Section 4.2, the normalized power excess (NPE) plays the role of the test statistic. The null hypothesis is that a 5 σ ALP signal is present; the alternative is that no signal is present. Monte Carlo simulations of the NPE distribution under the null hypothesis determine the threshold $\Theta_{\mathrm{RE}} = 3.43$ corresponding to the 95 % confidence level (see Section 4.2).

B.1.5 Gaussian example

For $X \sim \mathcal{N}(\mu, \sigma^2)$ with $H_0 : \mu = 0$ versus $H_1 : \mu = 1$, the log-likelihood ratio simplifies to $\log \Lambda(x) = x/\sigma^2 - 1/(2\sigma^2)$, so rejecting for large $\Lambda(x)$ is equivalent to rejecting for large x. This confirms that the sample mean is the optimal test statistic for Gaussian location testing.

C Optimal dwell time in frequency scanning

This appendix is adapted from Ref. [32].

This appendix derives the optimal dwell-time allocation scheme for a frequency scan, following the treatment of Ref. [32]. The goal is to maximize the sensitive area in the $\log \nu$–$\log |g_{\mathrm{aNN}}|$ parameter space over a total measurement time T_{tot}.

To simplify the notation, the frequency is normalized: $\hat{\nu} = \nu/\nu_{\mathrm{start}}$, with $\hat{\nu} \in [1,\, \hat{\nu}_{\mathrm{end}}]$ where $\hat{\nu}_{\mathrm{end}} = \nu_{\mathrm{end}}/\nu_{\mathrm{start}} > 1$. The dwell time is parameterized as

$$T = \kappa\, \hat{\nu}^{\,n}\,, \tag{C.1}$$

where κ is determined by T_{tot} and n, and the optimal value of n is determined below.

Regimes (1)–(3): step size proportional to ν

In regimes (1), (2), and (3), the frequency-step size is proportional to $\hat{\nu}$ (either one ALP linewidth $\hat{\nu}/Q_a$ or one NMR linewidth $\delta_\nu \hat{\nu}$). The scan rate is

$$\frac{d\hat{\nu}}{dt} = \frac{\Gamma_{\mathrm{scan}}}{T} \propto \hat{\nu}^{1-n}\,, \tag{C.2}$$

so that

$$dt = \kappa\, C\, \hat{\nu}^{n-1}\, d\hat{\nu}\,, \tag{C.3}$$

where $C = Q_a$ in regime (1) and $C = \delta_\nu^{-1}$ in regimes (2) and (3). Integrating over $[0,\, T_{\mathrm{tot}}]$ and $[1,\, \hat{\nu}_{\mathrm{end}}]$ gives

$$\kappa = \frac{n\, T_{\mathrm{tot}}}{C(\hat{\nu}_{\mathrm{end}}^{n} - 1)}\,, \qquad n \neq 0\,. \tag{C.4}$$

At $n = 0$: $\kappa|_{n=0} = T_{\mathrm{tot}}/[C\ln(\hat{\nu}_{\mathrm{end}})]$.

Regime (4): step size independent of ν

In regime (4) ($T_2^* = T_2 \ll \tau_a$), the step size is one NMR linewidth $2/T_2$, independent of frequency. The scan rate is proportional to $\hat{\nu}^{-n}$, giving

$$\kappa = \frac{2(n+1)\, T_{\text{tot}}}{(\hat{\nu}_{\text{end}}^{n+1} - 1)\, T_2}\,, \qquad n \neq -1\,. \tag{C.5}$$

At $n = -1$: $\kappa|_{n=-1} = 2T_{\text{tot}}/[\ln(\hat{\nu}_{\text{end}})\, T_2]$.

Optimal exponent n

The sensitive area in log-log scale is

$$A(n) = -\int_1^{\hat{\nu}_{\text{end}}} \log f(\hat{\nu},\, n)\, d \log \hat{\nu}\,, \tag{C.6}$$

where $f(\hat{\nu},\, n)$ is the exclusion threshold from Equation (3.10) with the dwell time from Equation (C.1). Taking the derivative $A'(n)$ and setting it to zero shows:

- For regimes (1), (2), and (3): $A'(n) > 0$ for $n < 0$ and $A'(n) < 0$ for $n > 0$, so the maximum is at $n = 0$ (uniform dwell time).
- For regime (4): the analysis shifts the effective exponent from n to $n + 1$, giving an optimum at $n = -1$ (dwell time $T \propto \hat{\nu}^{-1}$).

An intuitive restatement: when the sensitive bandwidth per step grows with frequency (regimes 1–3), the number of steps per unit $\log \nu$ is constant, and uniform dwell time is optimal. When the bandwidth is independent of frequency (regime 4), spending less time at higher frequencies—where the ALP coherence time is shorter and the signal integrates faster—maximizes the covered area.

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